# 7 The Application Layer

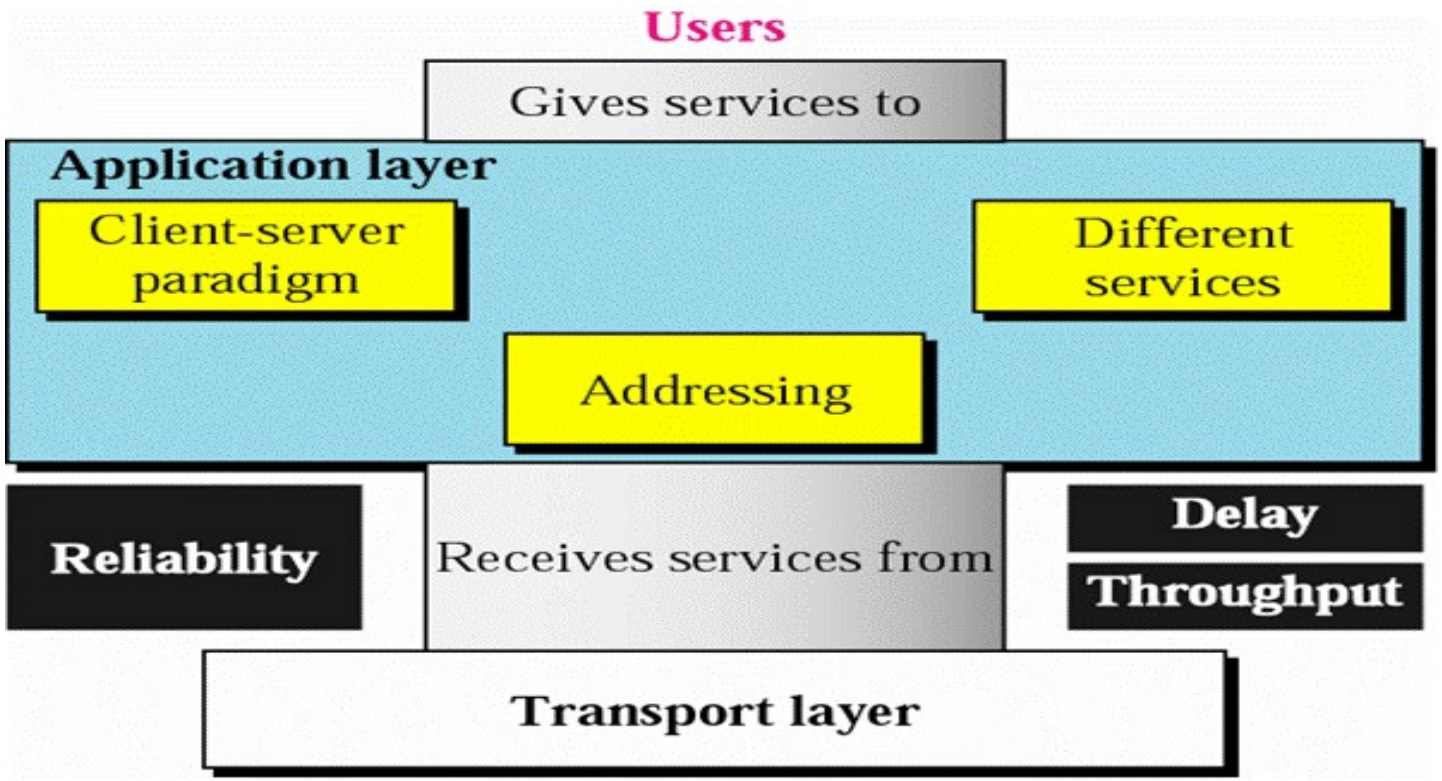
(about 3-4 hours)

# Outline

- *Introduction to Application Layer*
  - Positions and functions of application layer
  - Peer-to-peer paradigm vs. Client-server paradigm
- Domain Name System (DNS)
- Email System and Protocols
- World Wide Web (WWW)

# Position of Application Layer

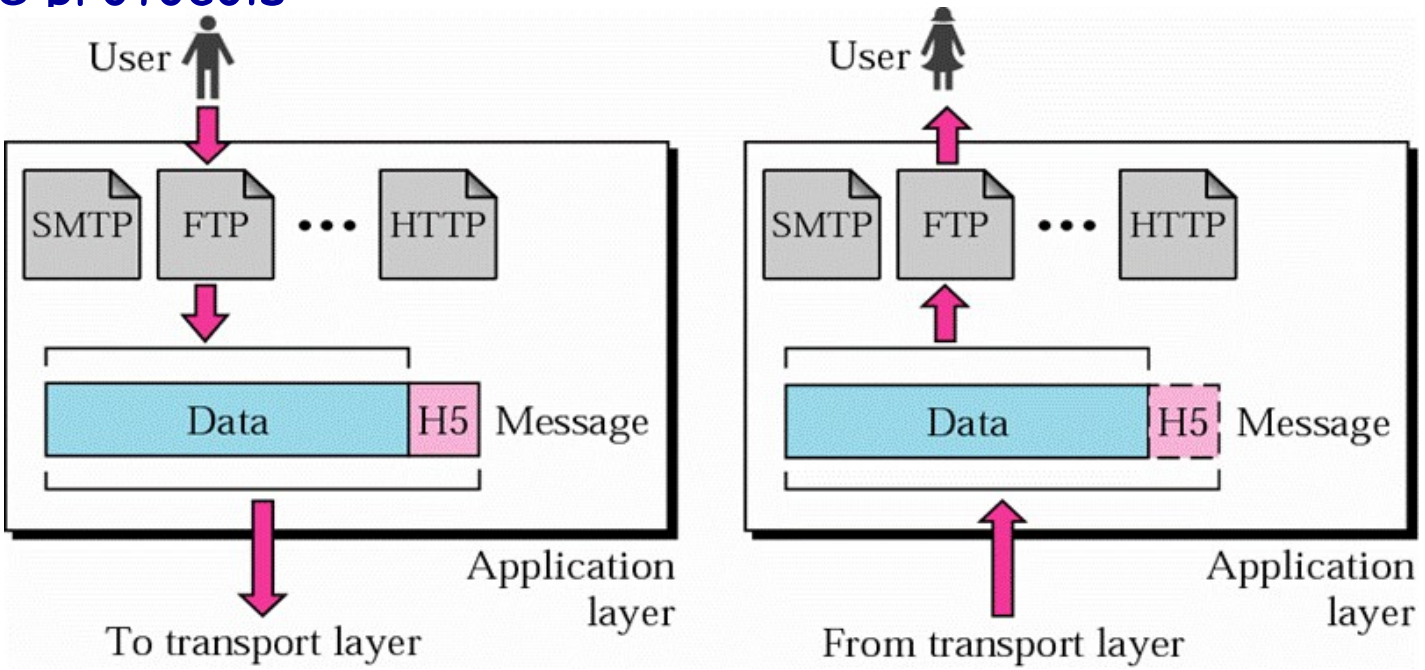
- Top layer of the Layering Architecture




*[Forouzan]*

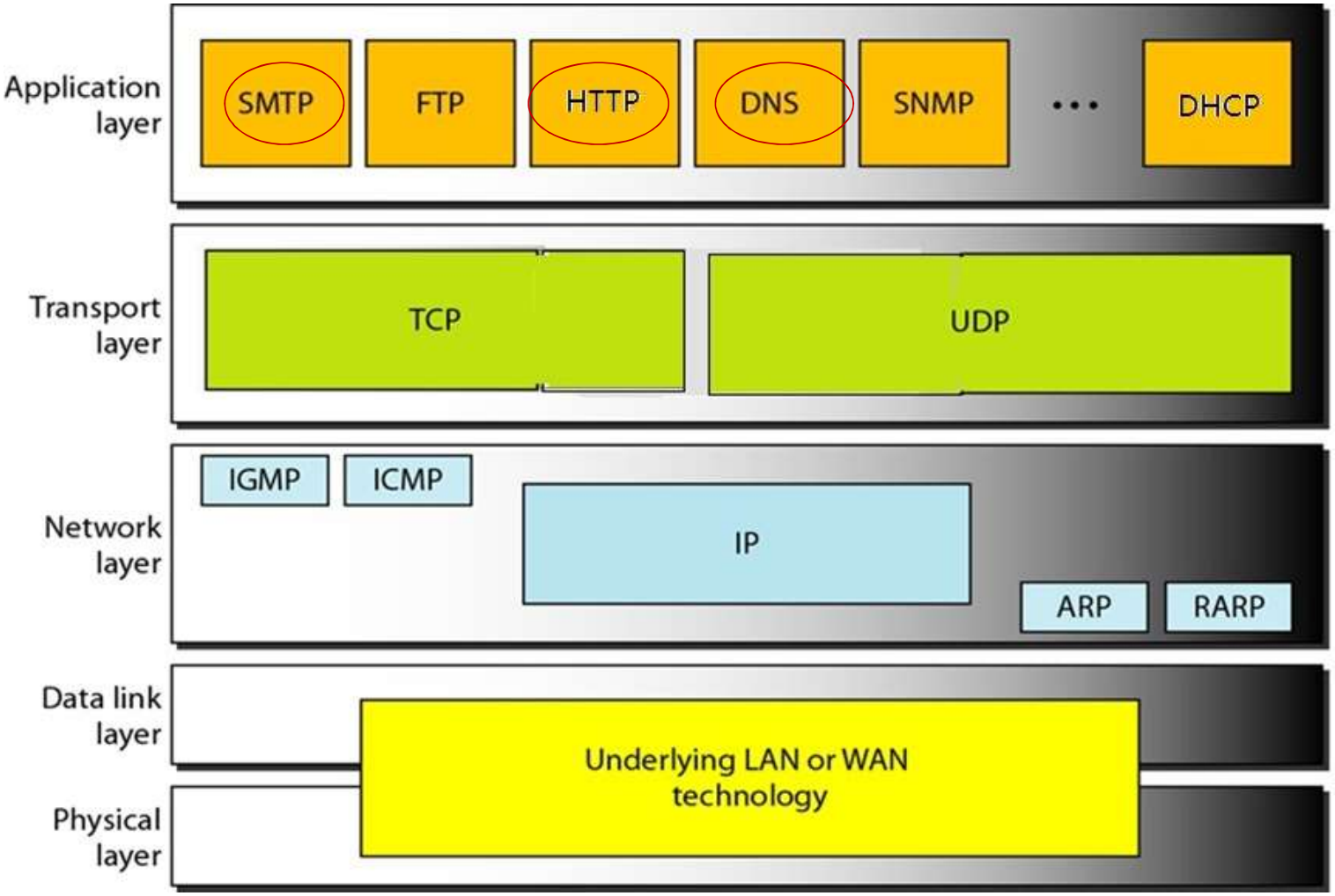
# Duties and Features of Application Layer

- Providing various applications to end users
- Maximum number of protocols and most complex
- Each protocol is specified to a type of application, no general-purpose protocols



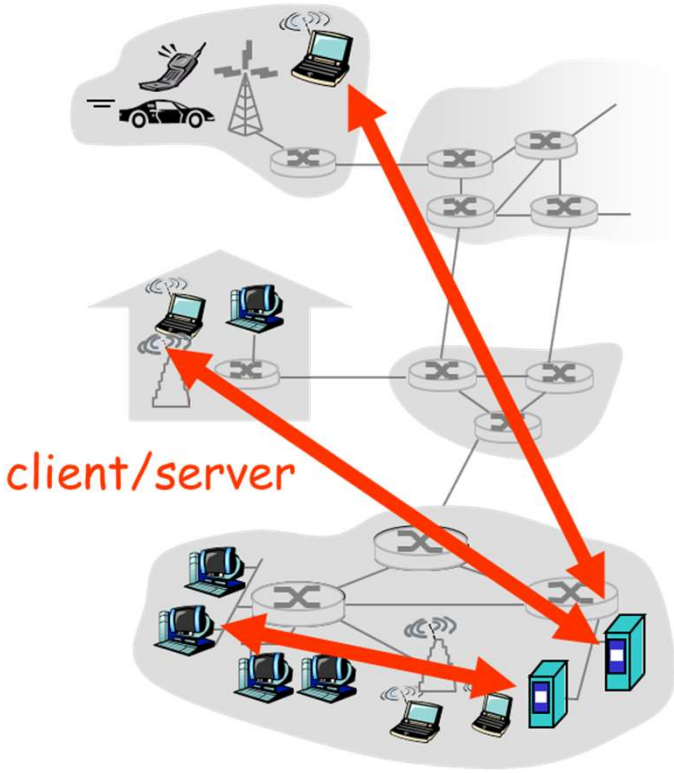

*[Forouzan]*

# Conventional Internet Application Protocols



# Client-server Paradigm

- The tasks, capabilities and protocol machines are different at each side.
- server:
  - always-on
  - permanent IP address
  - server farms for scaling
- clients:
  - communicate with server
  - may be intermittently connected
  - may have dynamic IP addresses
  - do not communicate directly with each other

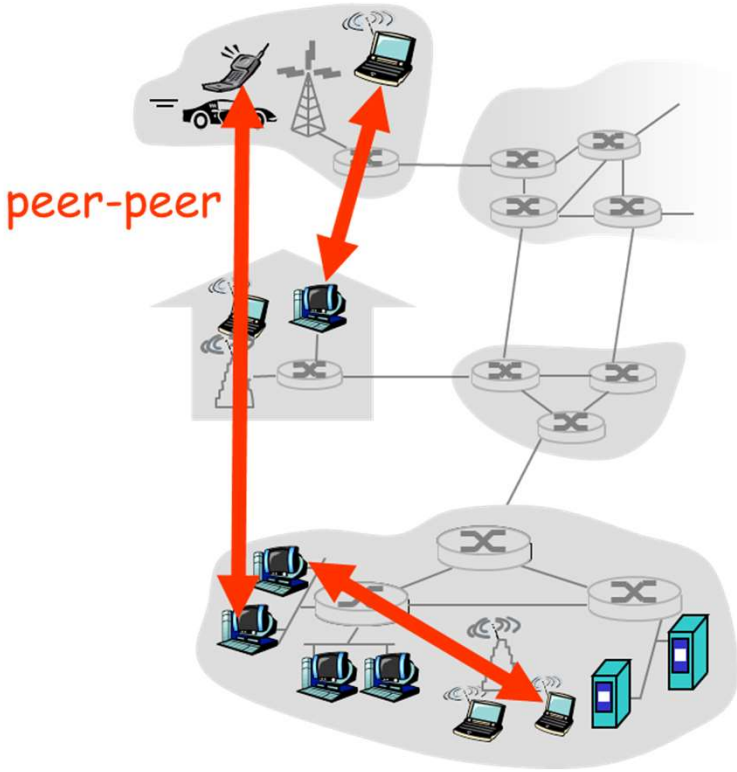



Forms the basis of computer networks

*[Kurose]*

# Pure P2P Architecture in Computer Networks

- *no* always-on server
- arbitrary end systems directly communicate
- peers are intermittently connected and change IP addresses




*[Kurose]*

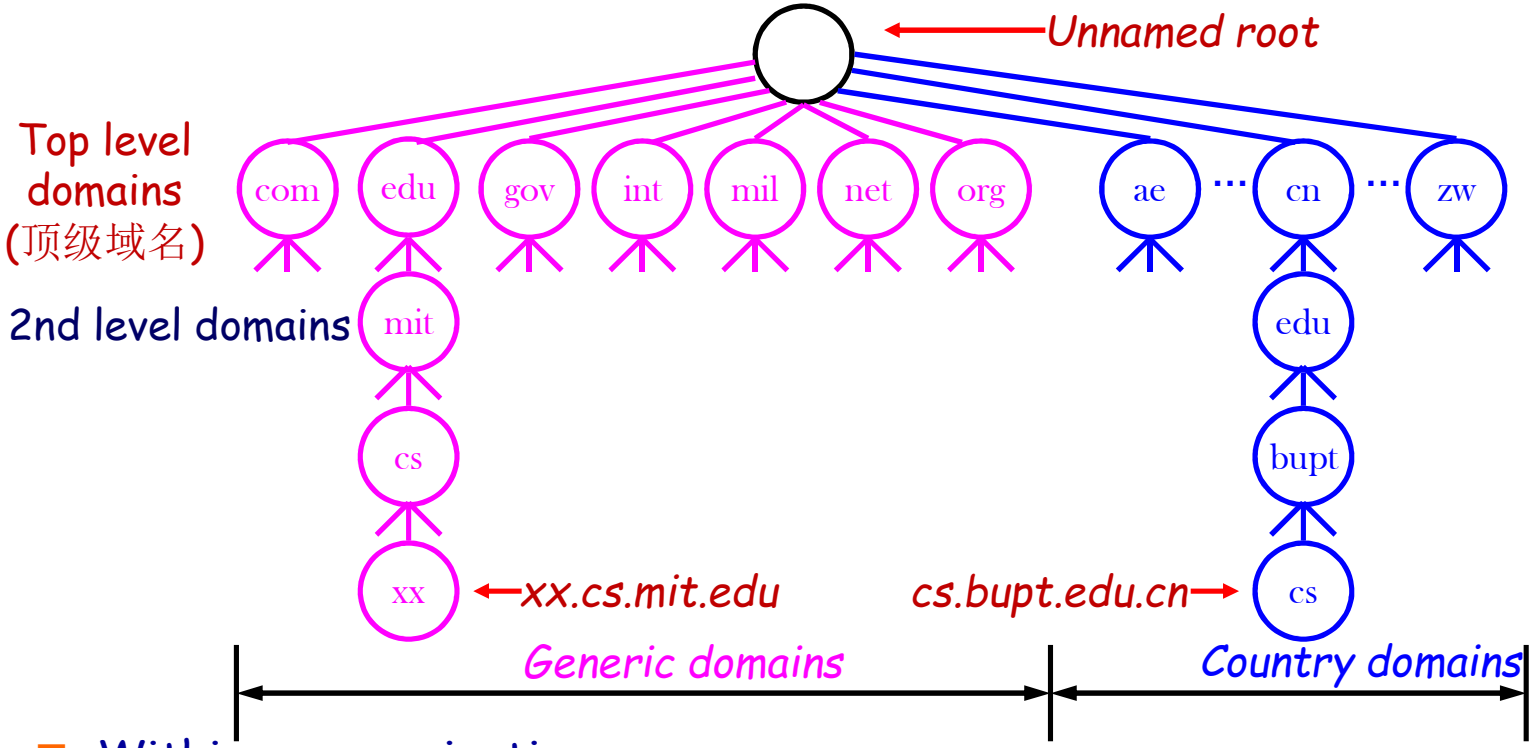
# Outline

- Introduction to Application Layer
- *Domain Name System (DNS) (Textbook 7.1)*
  - Hierarchical namespace
  - Resource records
  - Name servers
  - Name Resolution Procedure
- Email System and Protocols
- World Wide Web (WWW)

# DNS: Domain Name System(域名系统)

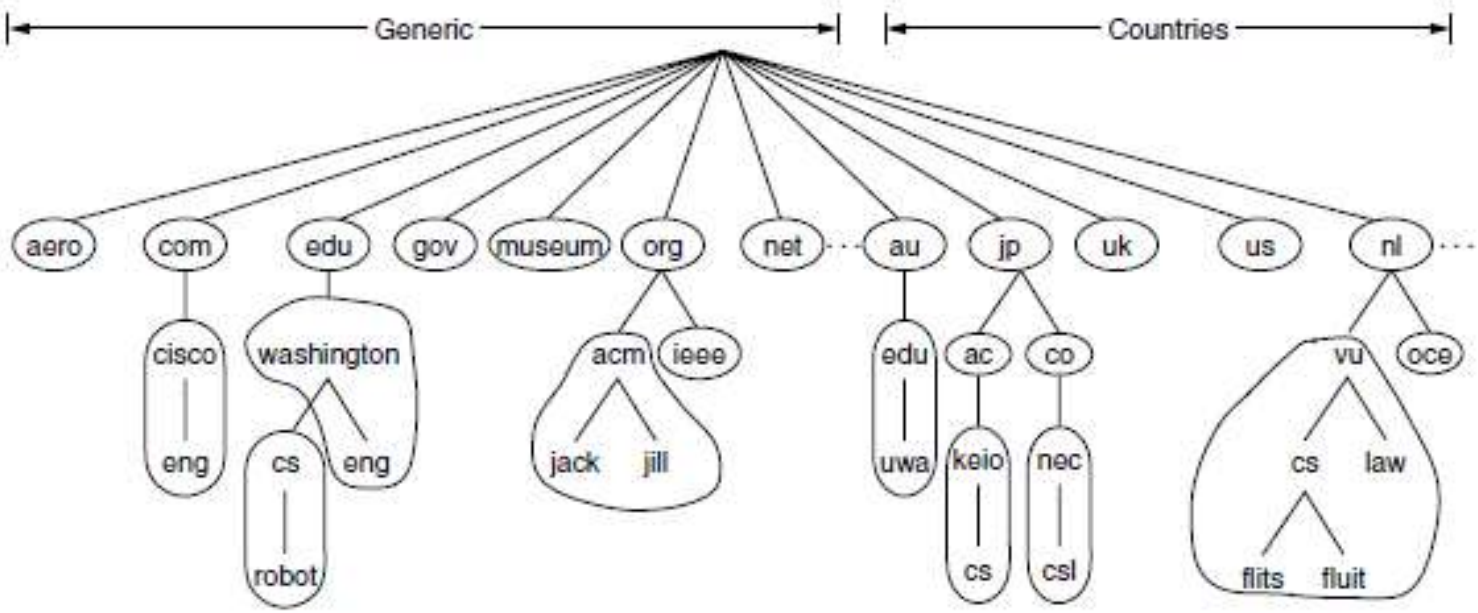
- A client-server application that identifies each host on the Internet with a unique user-friendly name(domain name),
  - e.g. www.bupt.edu.cn instead of 123.127.134.10
- Providing support for other applications
- Translating domain name (host name) to IP address
- Out-dated "hosts.txt"
- Features
  - *Hierarchical namespace*: Host have a composite names that are all hierarchically organized
    - In contrast to *Flat Name Space*: Each host in the network is unique and independent from each other
  - Distributed database

# Hierarchical Namespace




- Within an organization,
  - Subdivision possible
  - Arbitrary levels possible (maximum depth =128)
  - Not standardized, controlled locally by the organization
- Naming policy: by the path upward from leaf to root, separated by "."

# Zones(区)(Figure 7-5)




- A zone corresponds to an administrative authority that is responsible for that portion of the hierarchy
- Eg. bupt controls *x.bupt.edu.cn*

# Resource Records(资源记录)

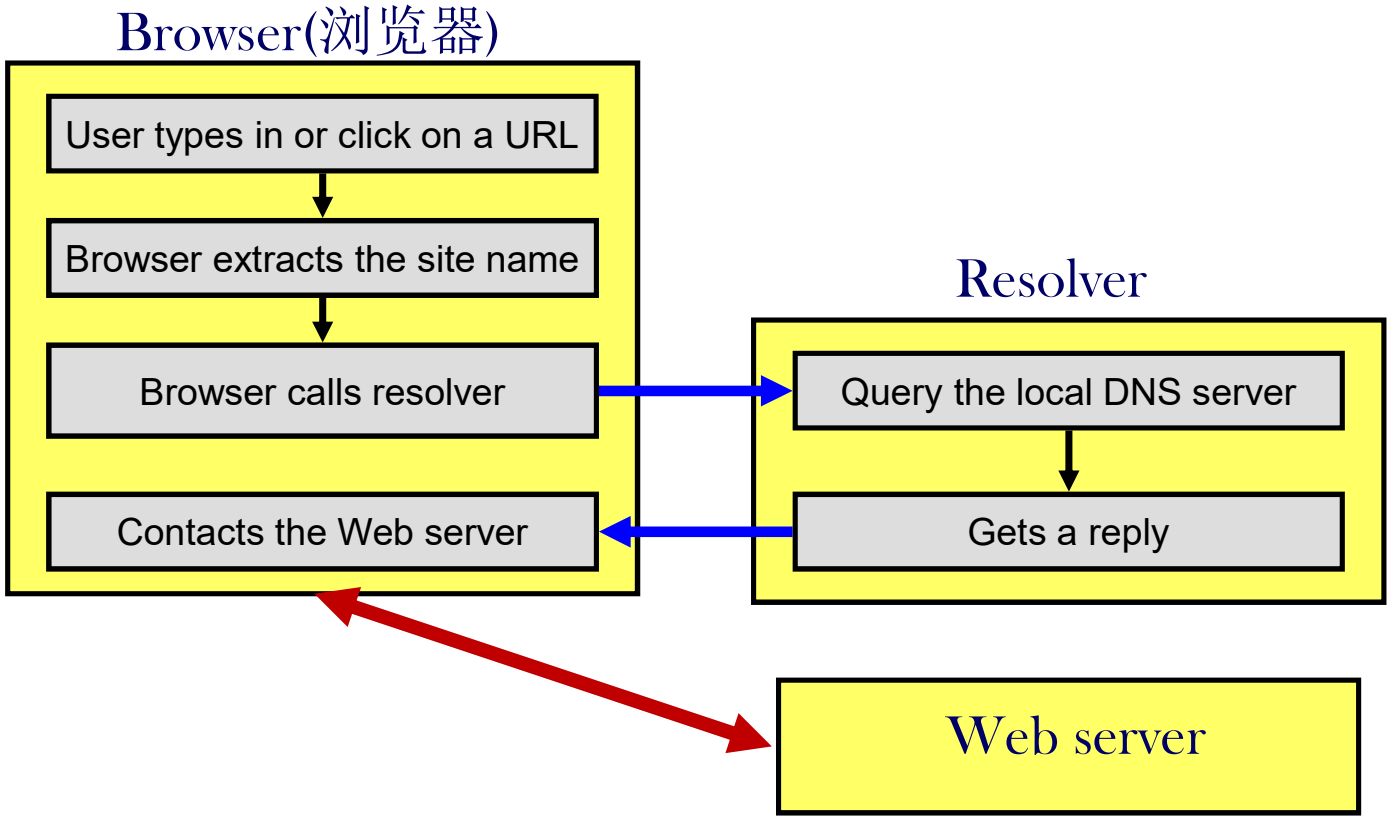
- Each domain in the DNS has one or more Resource Records (RRs)
- Each RR has the following information
  - Owner: the domain name
  - Type: specifies the type of the resource in this RR
    - SOA – Start Of Authority, identifies the zone
    - A – Host Address
    - MX – Mail Exchanger
    - CNAME – Canonical Name, real name used for management
    - …
  - Class: specifies the protocol family to use
    - IN – the Internet system
  - TTL: specifies the Time To Live (in unit of second) of the cached RRs

# Sample of DNS Database

| Domain name | Time-to-live | Class | Type | Value |
|---|---|---|---|---|
| ; Authoritative data for cs.vu.nl | | | | |
| cs.vu.nl. | 86400 | IN | SOA | star boss (9527,7200,7200,241920,86400) |
| cs.vu.nl. | 86400 | IN | MX | 1 zephyr |
| cs.vu.nl. | 86400 | IN | MX | 2 top |
| cs.vu.nl. | 86400 | IN | NS | star |
| | | | | |
| star | 86400 | IN | A | 130.37.56.205 |
| zephyr | 86400 | IN | A | 130.37.20.10 |
| top | 86400 | IN | A | 130.37.20.11 |
| www | 86400 | IN | CNAME | star.cs.vu.nl |
| ftp | 86400 | IN | CNAME | zephyr.cs.vu.nl |
| | | | | |
| flits | 86400 | IN | A | 130.37.16.112 |
| flits | 86400 | IN | A | 192.31.231.165 |
| flits | 86400 | IN | MX | 1 flits |
| flits | 86400 | IN | MX | 2 zephyr |
| flits | 86400 | IN | MX | 3 top |
| | | | | |
| rowboat | | IN | A | 130.37.56.201 |
| | | IN | MX | 1 rowboat |
| | | IN | MX | 2 zephyr |
| | | | | |
| little-sister | | IN | A | 130.37.62.23 |
| | | | | |
| laserjet | | IN | A | 192.31.231.216 |

# DNS Client

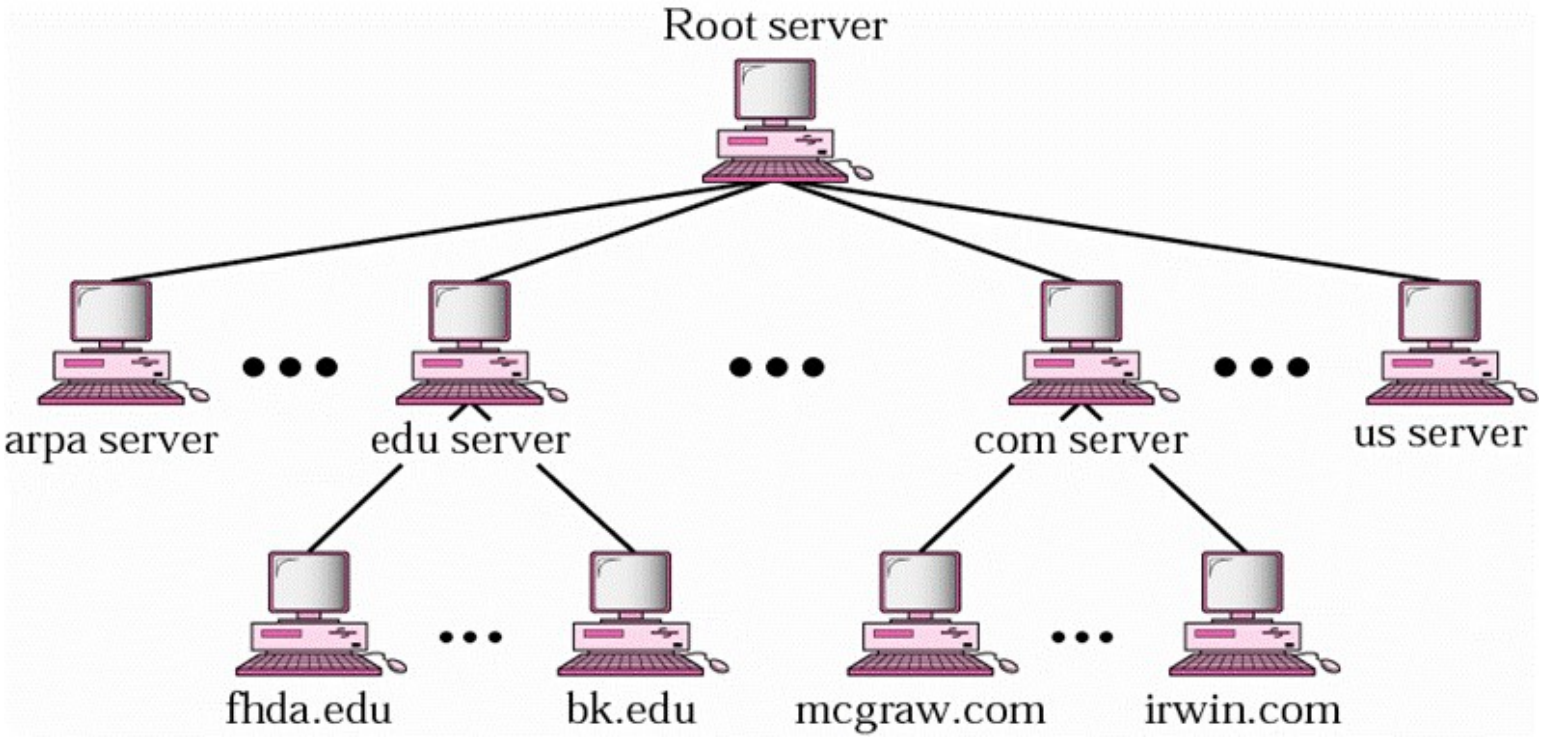
- Known as resolver(解析器)
  - software running on client, able to access at least one name server, i.e. Local name server(本地名字服务器)



# DNS Servers

- Multiple DNS servers used
  - Arranged in hierarchy
  - Each server has authority over a portion of the hierarchy: Authoritative Name Server(权威名字服务器)
  - Each server contains all the records for the hosts in its zone
  - How does a server know about other servers that are responsible for the other zones?
    - Every server knows the root
    - Root server knows about all TLD servers
    - Every server knows the server(s) further down the hierarchy

# Hierarchy of Name Servers




*[Forouzan]*

# Primary and Secondary Servers

- DNS uses backup server(s)
- Primary server: creates, maintains, and updates information about its zone
- Secondary server: gets its information from a primary server; as a backup server in case of failure
- Both servers have authority over their zone
- ISPs
  - Offering DNS service to subscribers
- Small organizations and individuals
  - Only need domain names for computers running servers
  - Contract with an ISP for domain service

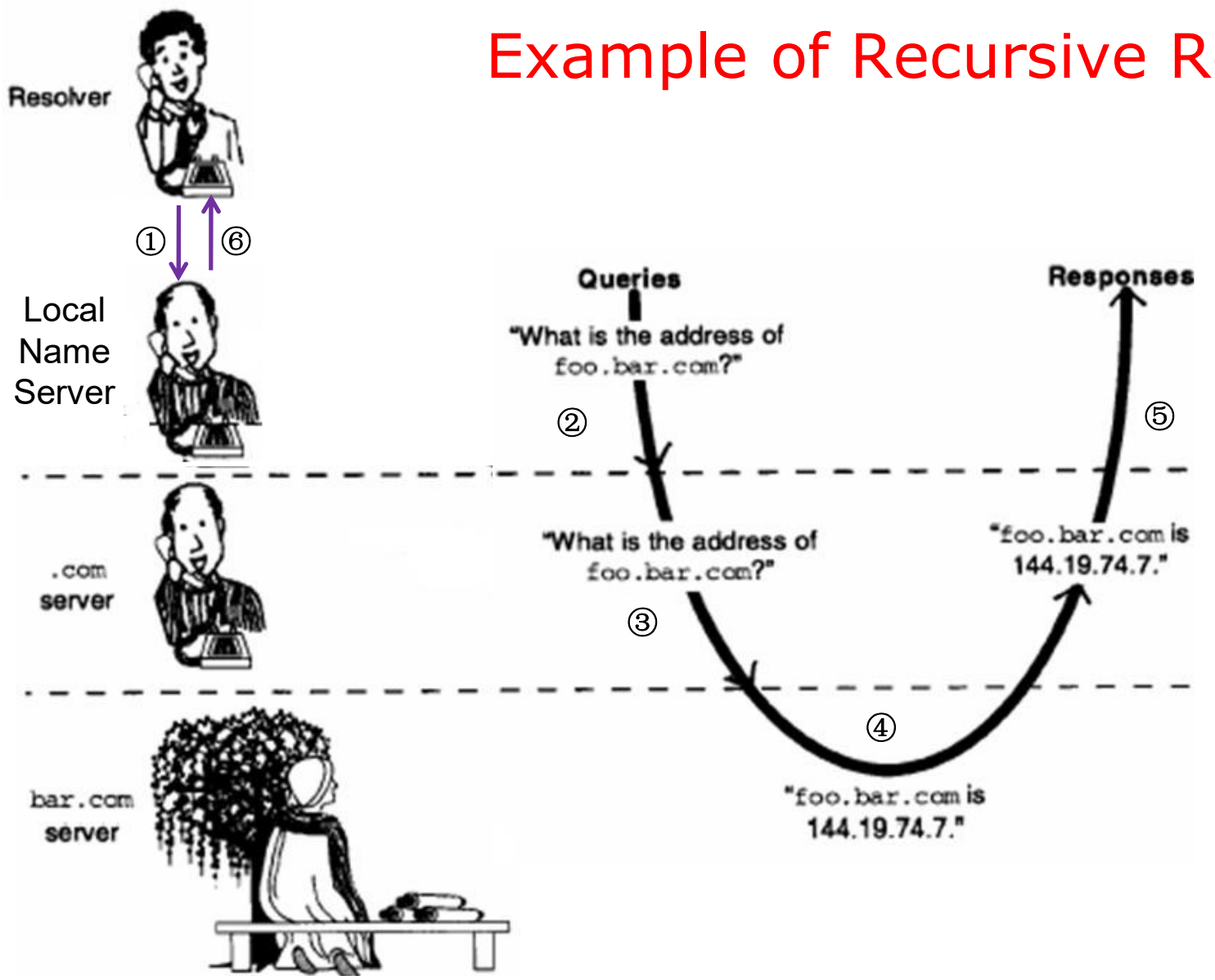
# DNS Lookup

- Applications
  - Becoming DNS client
  - Sending request first to local name server
- Local name server
  - Listening at port 53, normally using UDP
  - If answer known, returns response
  - If answer unknown
    - Requests root or TLD name server
    - Follows links
    - Until reaching Authoritative Name Server, which performs name translation and returns response

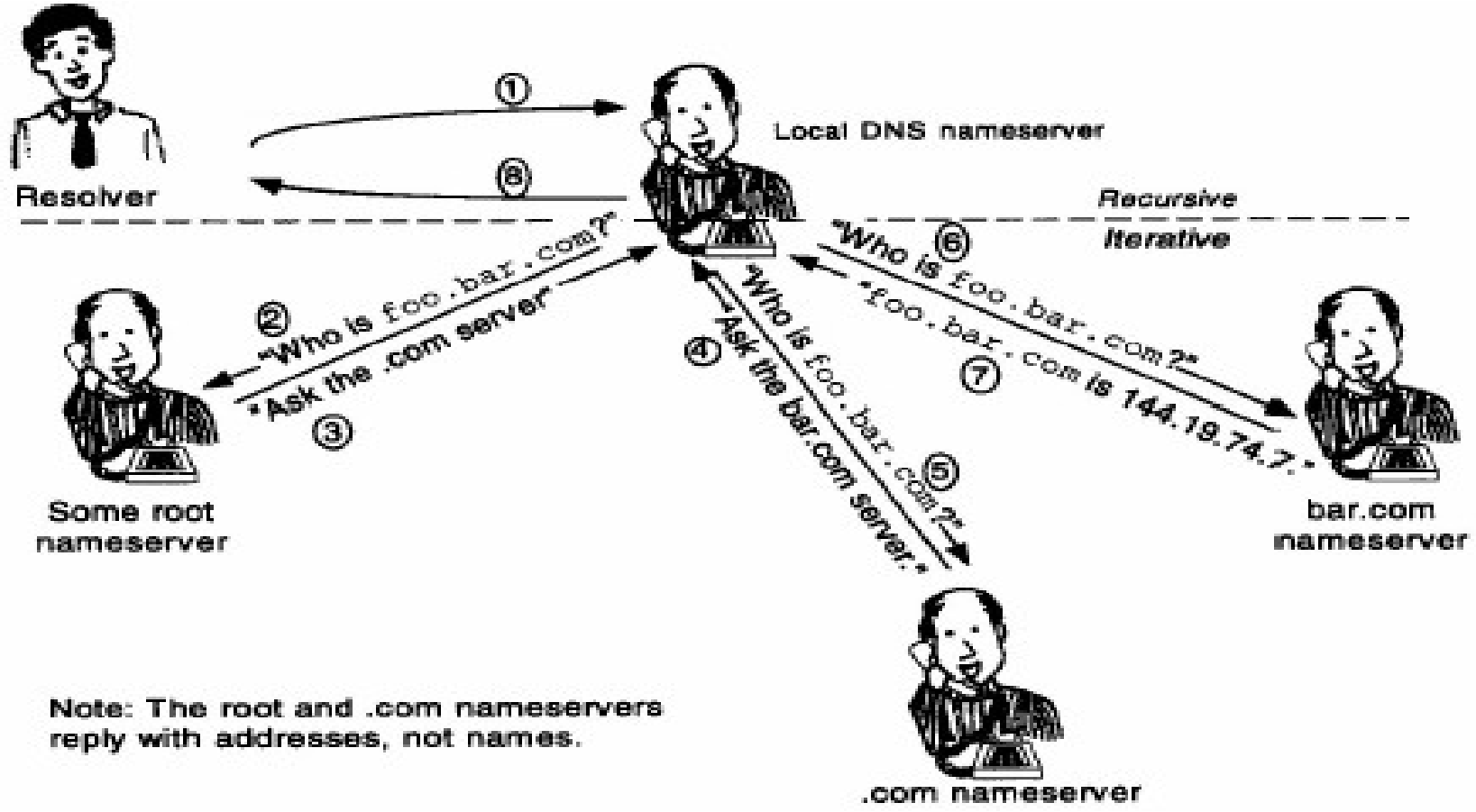
# Name Resolution Methods

- A query is made to a local name server. If local name server can not resolve the name, then it will query another server.
- Recursive resolution(递归解析/递归查询):
  - If the queried server does not have the information, it must make the appropriate query or queries to get the information
  - Generally, a server fulfills a recursive query either with data in its own memory or by making another recursive query
- Iterative resolution(迭代解析/迭代查询):
  - If the queried server does not have the information, it may then respond with the address of another server; the local name server (on behalf of resolver )then queries that server (which might respond with the address of another server, and so on)
  - Commonly used by name servers on the Internet

# Example of Recursive Resolution

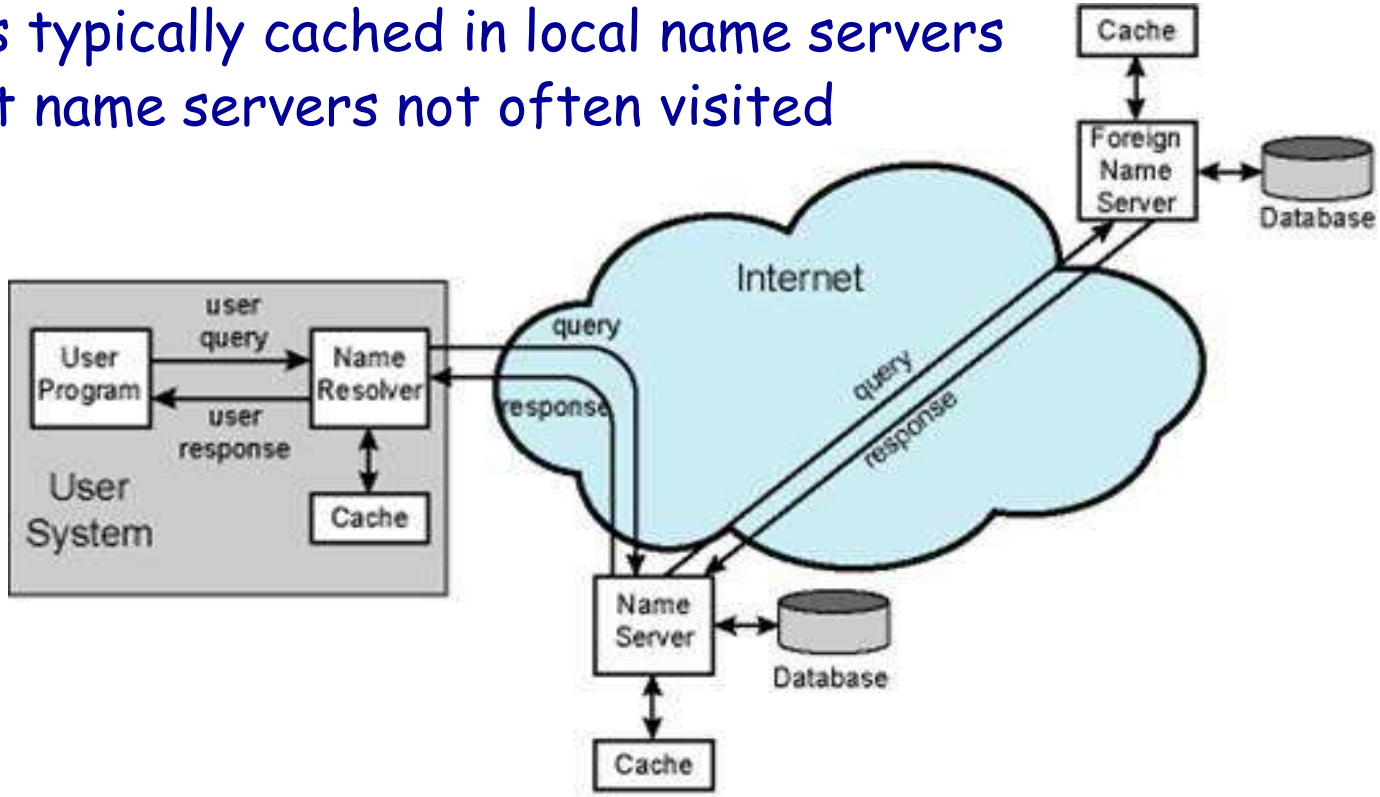


# Example of Iterative Resolution



# DNS Caching

- Server always caches answers
- Host can cache answers
- TLD servers typically cached in local name servers
  - Thus root name servers not often visited



# DNS Tools: nslookup

- Query name servers iteratively

```
C:\Users\Administrator>nslookup -query=MX bupt.edu.cn
服务器:  UnKnown
Address:  10.3.9.4

非权威应答:
bupt.edu.cn     MX preference = 10, mail exchanger = mxbiz2.qq.com
bupt.edu.cn     MX preference = 5, mail exchanger = mxbiz1.qq.com

C:\Users\Administrator>nslookup mxbiz2.qq.com
服务器:  UnKnown
Address:  10.3.9.4

非权威应答:
名称:    mxbiz2.qq.com
Addresses:  163.177.89.176
          61.241.49.98


C:\Users\Administrator>
```

# DNS tools: dig

```
student@BUPTIA:~$ dig www.bupt.edu.cn

; <<>> DiG 9.9.5-3ubuntu0.8-Ubuntu <<>> www.bupt.edu.cn
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 5439
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
;; QUESTION SECTION:
;www.bupt.edu.cn.               IN      A

;; ANSWER SECTION:
www.bupt.edu.cn.        2796    IN      CNAME   vn.bupt.edu.cn.
vn.bupt.edu.cn.         68      IN      A       10.3.9.161
```

# DNS小结

- 应用层：面向用户，每个协议支持一类应用
- 两种典型的应用模式：C/S模式和P2P模式
- DNS目标：实现域名和IP地址之间的映射(转换)
  - 分层的域名空间、分布式数据库
  - 域名的规定
  - Client(解析器)请求本地域名服务器（UDP， 端口53）
  - 本地名字服务器请求其他服务器(最终到权威域名服务器)
    - 迭代解析（因特网普遍采用）
    - 递归解析
- DNS目标很简单，但实现系统庞大而复杂
- 具有高效率(缓存机制)和可靠性(镜像、辅助域名服务器)
- 安全性：DNSSEC

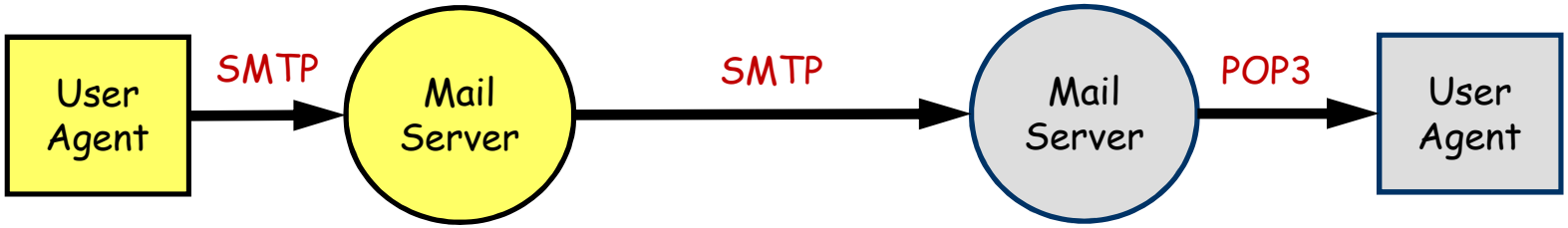
# Outline

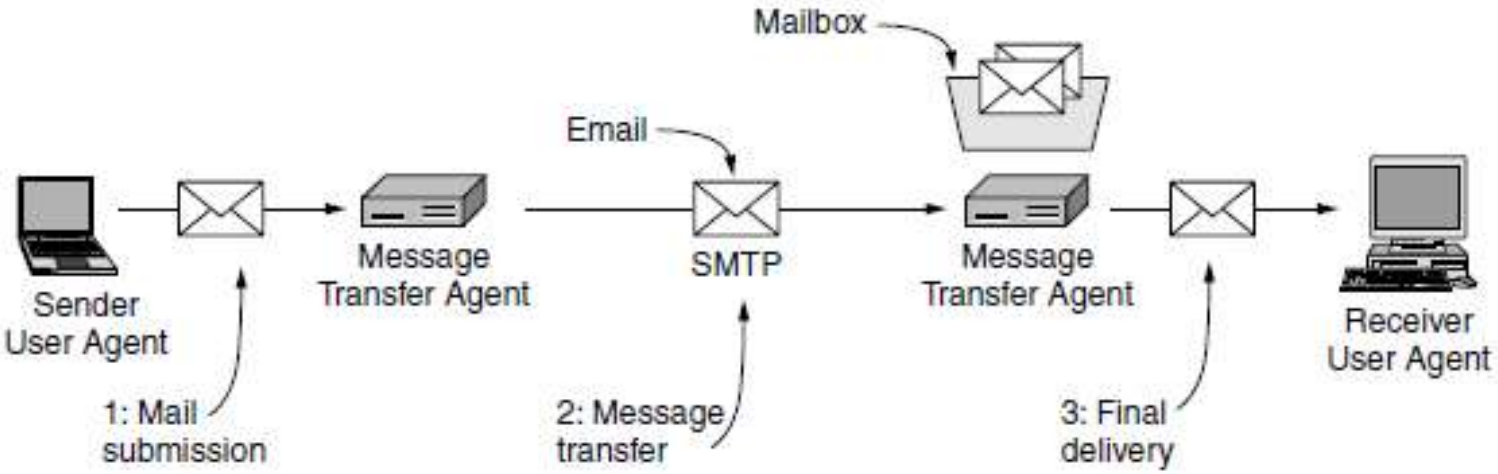
- Introduction to Application Layer
- Domain Name System (DNS)
- *Email System and Protocols*
  - System components
  - Message Format
  - SMTP
  - POP3
- World Wide Web (WWW)

# What is Email?

- Electronic Mail (email, e-mail)
- Provides a means to send electronic messages from one person to another asynchronously
- One of the most popular applications on the Internet and one of the most important communication methods today
- Email service can be provided by
  - ISPs: @126.com, @163.com, @sina.com, yahoo.com.cn, …
  - Corporations and institutes: @baidu.com, @bupt.edu.cn, @ietf.org, …
  - Bundled with other services: @139.com, @qq.com, …

# Components Of Email System (1)



# Components Of Email System (2)

- UA (User Agent)
  - end-user mail program
  - Interface between the end users and the email servers
  - E.g., outlook, foxmail, …
- Mail Server
  - Responsible for transmitting/receiving emails and reporting status information about mail transferring to the mail sender
  - Both a client and a server
- Email protocols
  - SMTP: used for sending an email
  - POP3: used for receiving an email

# Basic Functions Of Email System

- Composition - refers to the process of creating messages and answers.
- Transfer - refers to moving messages from the originator to the recipient.
- Reporting - refers to the process of informing the originator what happened to the message.
- Displaying - means of showing the messages.
- Disposition - refers to what happened to the message after it has been read by the receiver.

# Email Address

- The destination address must be in a format that the user agent can deal with
- Many user agents expect DNS addresses of the form *mailboxname@domain* ( RFC 5322 )
- Each email address is unique on the Internet because
  - *Domain* name is unique on the Internet
  - *Mailboxname* is unique in the domain

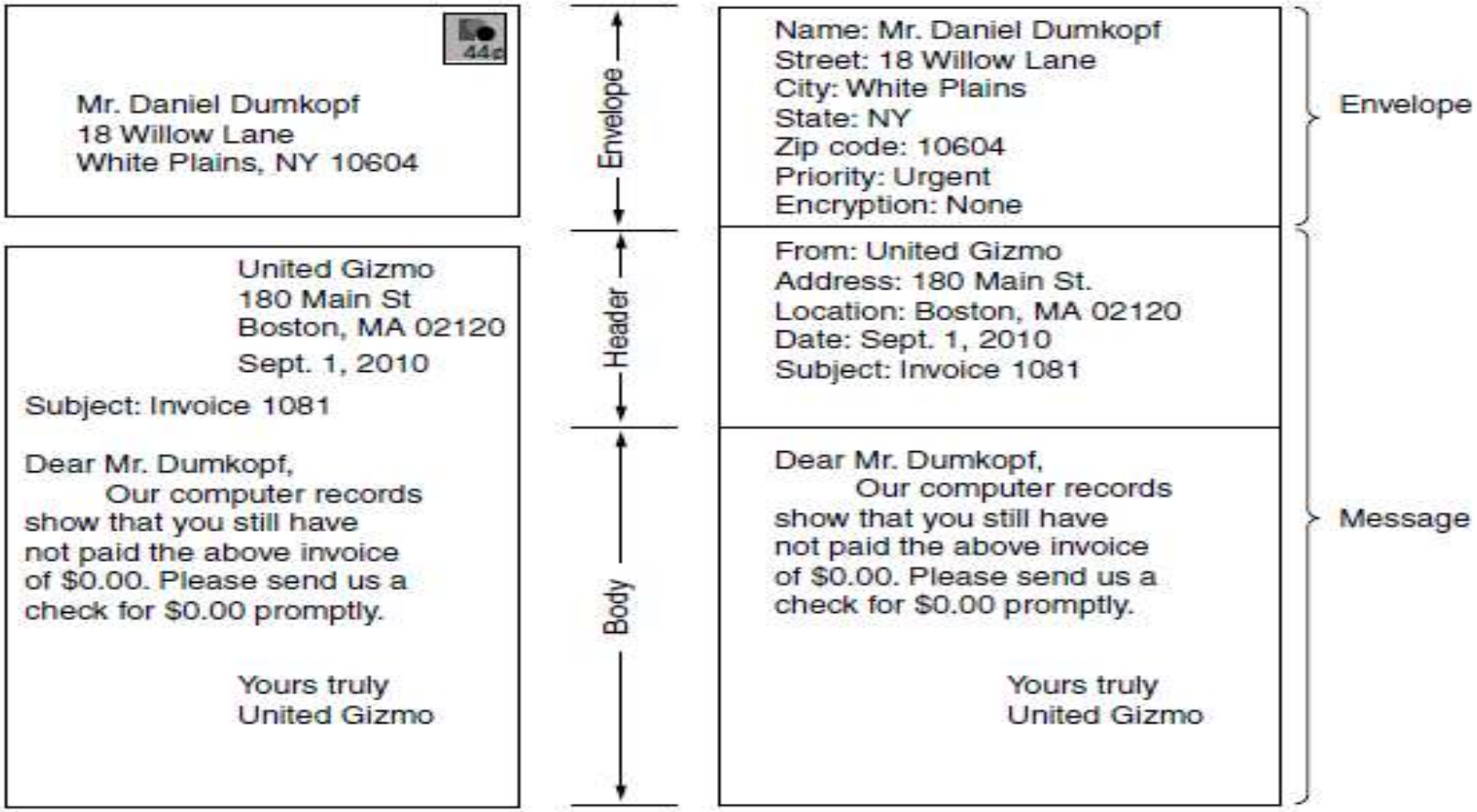
# Internet Message Format(RFC 5322)

- Message envelop
  - Contains whatever information is needed to accomplish transmission and delivery
  - Constructed by Mail Servers
- Message contents: comprise the object to be delivered to the recipient
  - Headers: from, to, subject, date, postmarks
  - Blank line
  - Body: actual message, may have many parts

# Message Headers Related to Message Transport

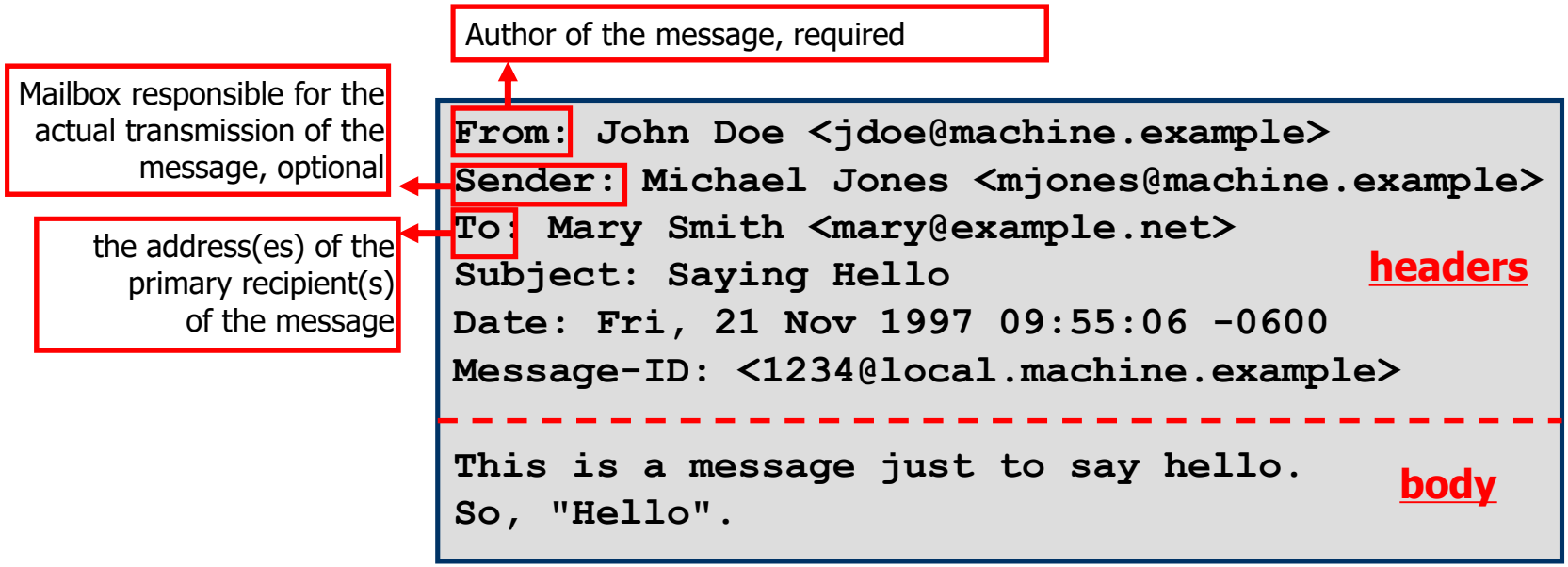
| Header | Meaning |
|---|---|
| To: | Email address(es) of primary recipient(s) |
| Cc: | Email address(es) of secondary recipient(s) |
| Bcc: | Email address(es) for blind carbon copies |
| From: | Person or people who created the message |
| Sender: | Email address of the actual sender |
| Received: | Line added by each transfer agent along the route |
| Return-Path: | Can be used to identify a path back to the sender |

# Paper Mail vs. Email



# Internet Message Format ——Example RFC 5322 Message

- The header part is everything up to the blank line, and the body is everything after the blank line




- Problem of RFC 5322: Only 7 bit ASCII format can be used

# MIME

- Multipurpose Internet Mail Extension
- Extension for multipart & multimedia email
- Additional mail headers define content
  - Type: text, image, audio, video, application and subtype within (eg text/html, image/gif)
  - Encoding (ASCII , quoted printable, base64) to handle arbitrary binary data when email system can only handle normal ASCII characters
- Supports multipart message content type
  - each part has its own type and encoding
- MIME messages can be sent using the existing mail programs and protocols
- Widely used now

# MIME: New Headers

| Header | Meaning |
|---|---|
| MIME-Version: | Identifies the MIME version |
| Content-Description: | Human-readable string telling what is in the message |
| Content-Id: | Unique identifier |
| Content-Transfer-Encoding: | How the body is wrapped for transmission |
| Content-Type: | Type and format of the content |

# MIME: Content Types and Subtypes

| Type | Subtype | Description |
|---|---|---|
| Text | Plain | Unformatted text |
| | Enriched | Text including simple formatting commands |
| Image | Gif | Still picture in GIF format |
| | Jpeg | Still picture in JPEG format |
| Audio | Basic | Audible sound |
| Video | Mpeg | Movie in MPEG format |
| Application | Octet-stream | An uninterpreted byte sequence |
| | Postscript | A printable document in PostScript |
| Message | Rfc822 | A MIME RFC 822 message |
| | Partial | Message has been split for transmission |
| | External-body | Message itself must be fetched over the net |
| Multipart | Mixed | Independent parts in the specified order |
| | Alternative | Same message in different formats |
| | Parallel | Parts must be viewed simultaneously |
| | Digest | Each part is a complete RFC 822 message |

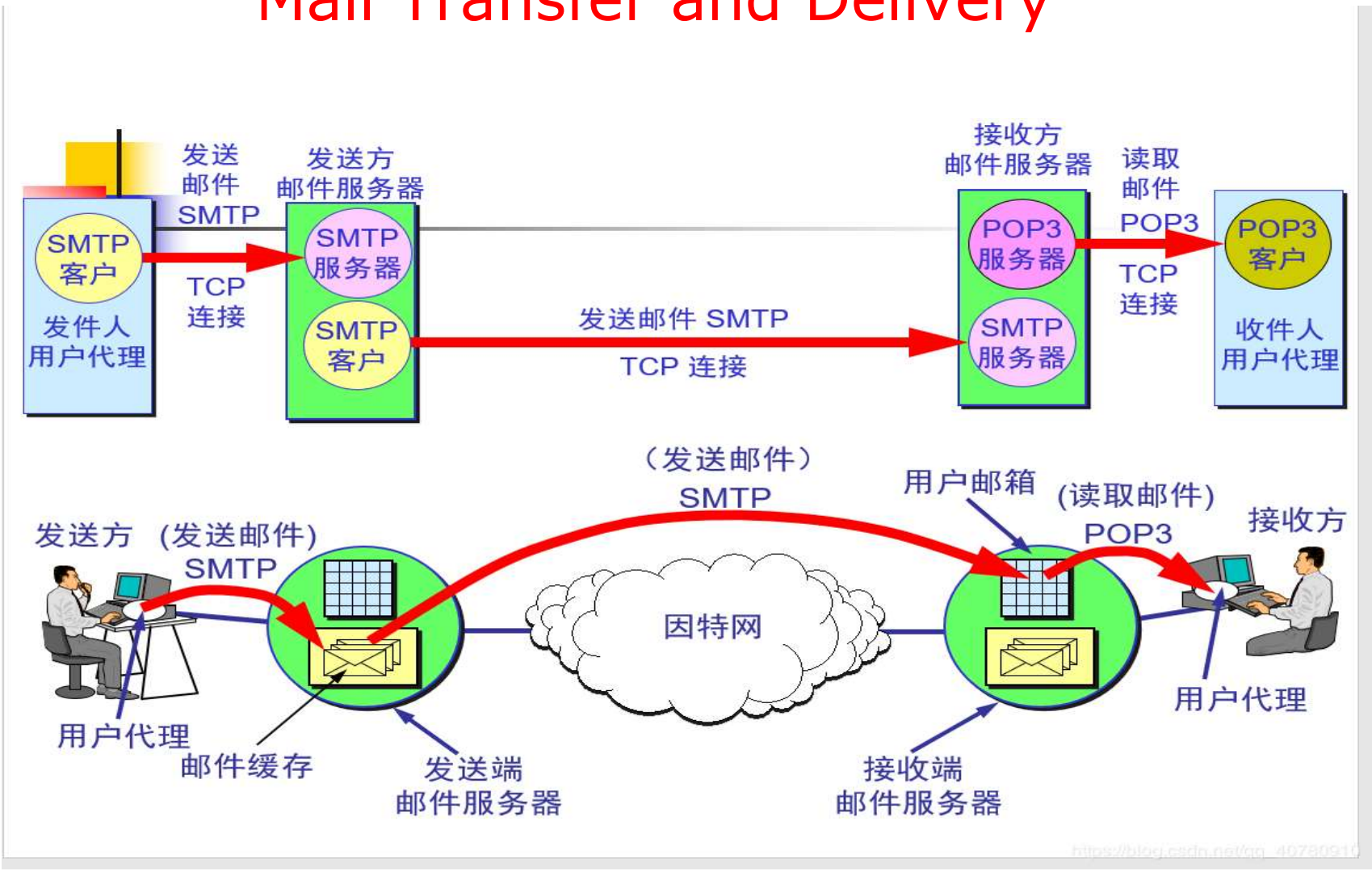
# Example of a Multipart Message(1)

```
From: alice@cs.washington.edu
To: bob@ee.uwa.edu.au
MIME-Version: 1.0
Message-Id: <0704760941.AA00747@cs.washington.edu>
Content-Type: multipart/alternative; boundary=qwertyuiopasdfghjklzxcvbnm
Subject: Earth orbits sun integral number of times

This is the preamble. The user agent ignores it. Have a nice day.

--qwertyuiopasdfghjklzxcvbnm
Content-Type: text/html

<p>Happy birthday to you<br>
Happy birthday to you<br>
Happy birthday dear <b> Bob </b><br>
Happy birthday to you</p>

--qwertyuiopasdfghjklzxcvbnm
Content-Type: message/external-body;
        access-type="anon-ftp";
        site="bicycle.cs.washington.edu";
        directory="pub";
        name="birthday.snd"

content-type: audio/basic
content-transfer-encoding: base64
--qwertyuiopasdfghjklzxcvbnm--
```

# Example of a Multipart Message(2)

- Saving the above message in *.eml and opening with Foxmail

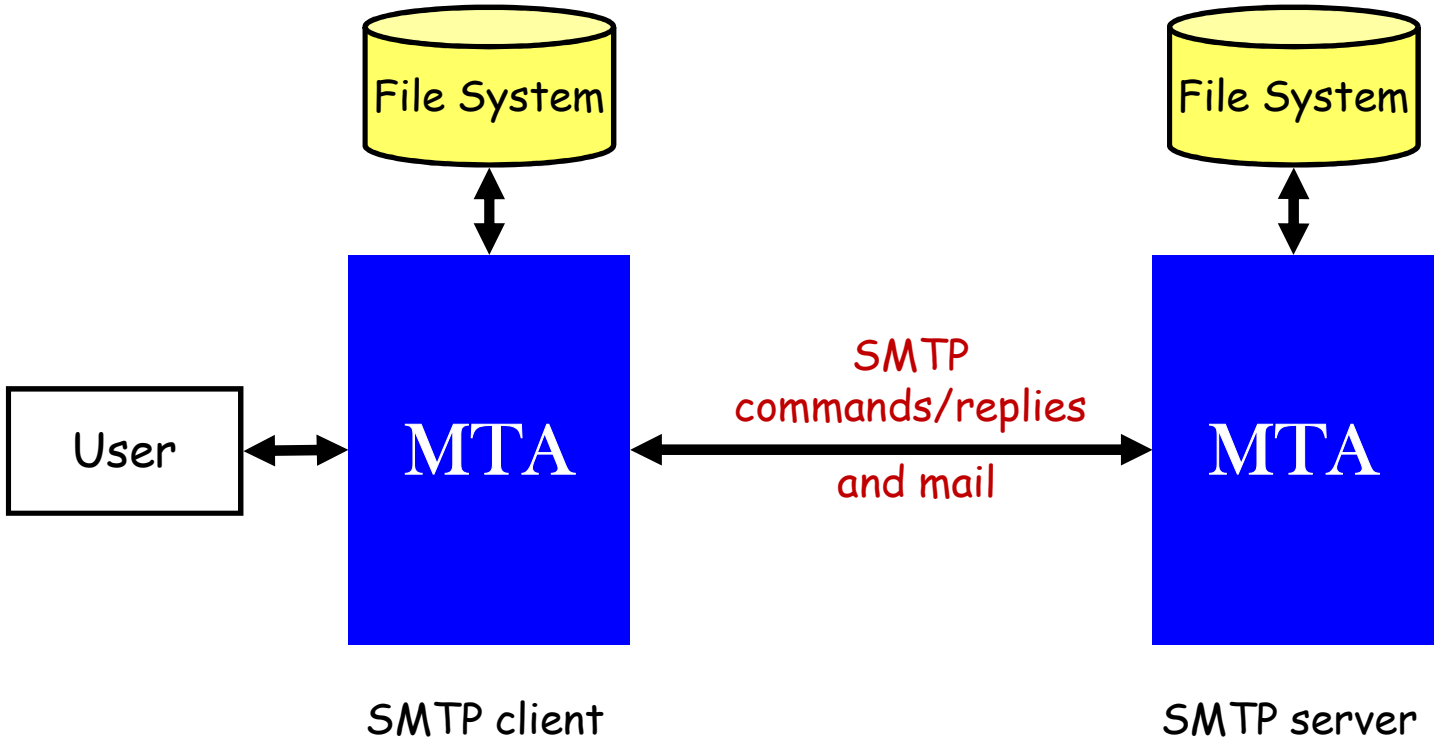
# Mail Transfer and Delivery



# SMTP

- Simple Mail Transfer Protocol
- Used in transferring mail messages from UA to mail server, and between mail servers
- SMTP server listens to port 25
- SMTP client establishes a TCP connection to this port
- If the message cannot be delivered, an error report containing the first part of the undeliverable message is returned to the sender – eventually
- SMTP is a simple ASCII protocol

# SMTP Basic Model



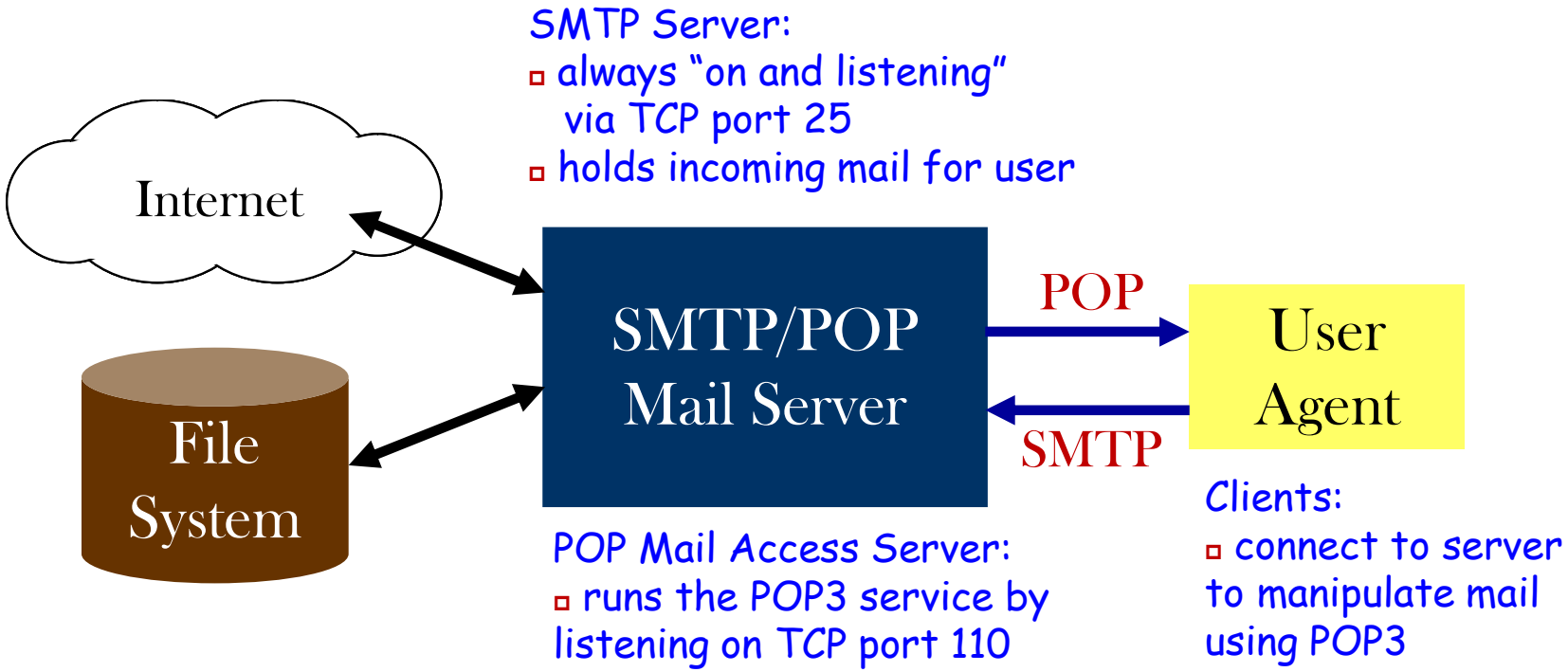

- MTA: Mail Transfer Agent, the mail server using SMTP

- Example of SMTP messages

```
                                S: 220 ee.uwa.edu.au SMTP service ready
C: HELO abcd.com
                                S: 250 cs.washington.edu says hello to ee.uwa.edu.au
C: MAIL FROM: <alice@cs.washington.edu>
                                S: 250 sender ok
C: RCPT TO: <bob@ee.uwa.edu.au>
                                S: 250 recipient ok
C: DATA
                                S: 354 Send mail; end with "." on a line by itself
C: From: alice@cs.washington.edu
C: To: bob@ee.uwa.edu.au
C: MIME-Version: 1.0
C: Message-Id: <0704760941.AA00747@ee.uwa.edu.au>
C: Content-Type: multipart/alternative; boundary=qwertyuiopasdfghjklzxcvbnm
C: Subject: Earth orbits sun integral number of times
C:
C: This is the preamble. The user agent ignores it. Have a nice day.
C:
C: --qwertyuiopasdfghjklzxcvbnm
C: Content-Type: text/html
C:
C: <p>Happy birthday to you
C: Happy birthday to you
C: Happy birthday dear <bold> Bob </bold>
C: Happy birthday to you
C:
C: --qwertyuiopasdfghjklzxcvbnm
C: Content-Type: message/external-body;
C:          access-type="anon-ftp";
C:          site="bicycle.cs.washington.edu";
C:          directory="pub";
C:          name="birthday.snd"
C:
C: content-type: audio/basic
C: content-transfer-encoding: base64
C: --qwertyuiopasdfghjklzxcvbnm
C: .
                                S: 250 message accepted
C: QUIT
                                S: 221 ee.uwa.edu.au closing connection
```

# POP: Basic Model

- Used to transfer mail from a mail server to a UA



# POP: Features

- Essentially store and forward.
  - Mail is stored on the server until the client connects and then is downloaded to the client.
  - You MAY be able to leave a copy on the server
- Simple protocol and widely used
  - Similar to SMTP command/reply lockstep protocol
  - Used to retrieve mail for a single user, requires authentication
- Many clients available such as foxmail, outlook
- Common used version: POP3 (POP Version 3)
- However, very bad for mobile users or users that use multiple machines during the day
  - Solution: IMAP (Internet Message Access Protocol)

# POP3: Example

```
                    S: +OK POP3 server ready
C: USER carolyn
                    S: +OK
C: PASS vegetables
                    S: +OK login successful
C: LIST
                    S: 1 2505
                    S: 2 14302
                    S: 3 8122
                    S: .
C: RETR 1
                    S: (sends message 1)
C: DELE 1
C: RETR 2
                    S: (sends message 2)
C: DELE 2
C: RETR 3
                    S: (sends message 3)
C: DELE 3
C: QUIT
                    S: +OK POP3 server disconnecting
```

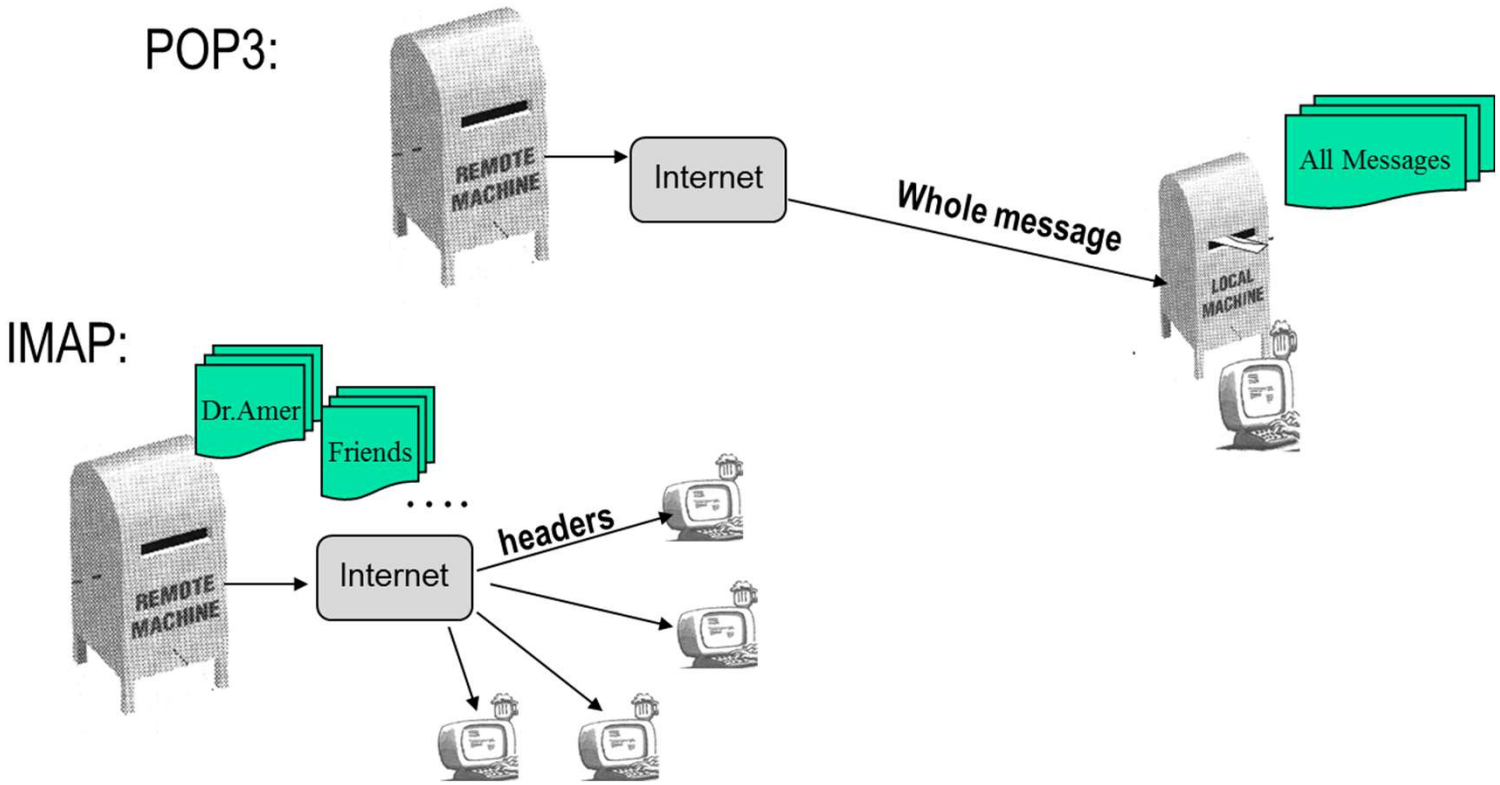
# IMAP (Internet Message Access Protocol)

- Features
  - Folders and messages can be stored either on the server or on the local computer
  - Since mails can remain on server, it is possible to access your same mail store even using a dumb terminal.
  - Much better for mobile users than POP
  - Can selectively copy (part of ) messages from the server to the local client based on many criteria
- QQ/163/gmail supports both IMAP and POP3
- Comparison of POP3 and IMAP
  - https://www.diffen.com/difference/IMAP_vs_POP3



48

# POP3 vs IMAP(1)

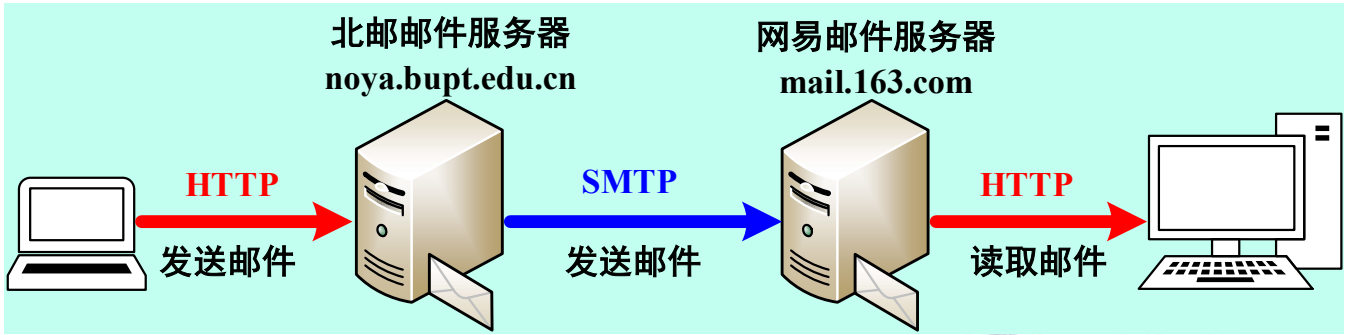


# POP3 vs IMAP

| Feature | POP3 | IMAP |
|---|---|---|
| Where is protocol defined? | RFC 1939 | RFC 2060 |
| Which TCP port is used? | 110 | 143 |
| Where is e-mail stored? | User's PC | Server |
| Where is e-mail read? | Off-line | On-line |
| Connect time required? | Little | Much |
| Use of server resources? | Minimal | Extensive |
| Multiple mailboxes? | No | Yes |
| Who backs up mailboxes? | User | ISP |
| Good for mobile users? | No | Yes |
| User control over downloading? | Little | Great |
| Partial message downloads? | No | Yes |
| Are disk quotas a problem? | No | Could be in time |
| Simple to implement? | Yes | No |
| Widespread support? | Yes | Growing |

# Web-based Mail: HTTP

- Can deliver mail message in web page format




北京郵電大學 厚德博学 敬业乐群

http://mail.bupt.edu.cn

# Email系统小结

- 目标：用户之间的异步数据传输
- 邮件格式
  - 基本邮件格式(IMF)只支持ASCII（英文）
  - MIME：扩展邮件头，支持多种数据类型、多种语言、邮件内包含多部分(附件)
- 邮件传输协议(Client/Server)
  - 发送邮件协议：SMTP
  - 接收邮件协议：POP3, IMAP4(功能更强，更复杂)
- Webmail
  - 用户使用浏览器和HTTP协议与邮件服务器通信，收发邮件
  - 邮件服务器之间使用SMTP发送邮件
- 安全性：PGP, SMIME

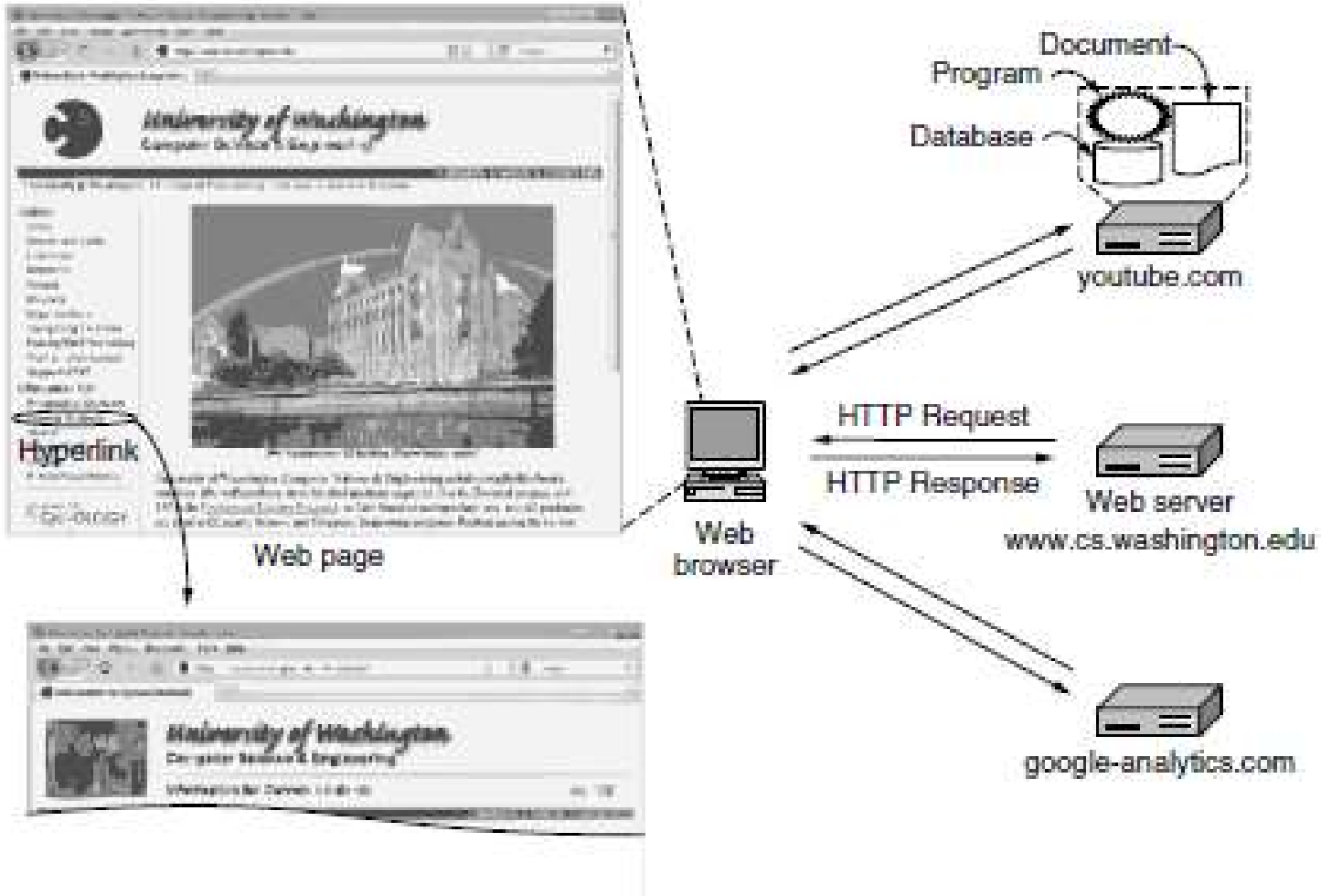
# Outline

- Introduction to Application Layer
- Domain Name System (DNS)
- Email System and Protocols
- *World Wide Web (WWW) (Textbook 7.3)*
  - WWW Architecture
  - Hypertext Transfer Protocol (HTTP)

# WWW: World Wide Web

- One of mostly widely used Internet application
- Not a network, but an "universe of network-accessible information, an embodiment of human knowledge" *(Tim Berners-Lee )*
- A distributed hyper-media system, including multiple media information such as text, graph, image, audio, even video
- Information is distributed in Internet, stored in web servers and accessed by hyperlinks(超级链接)
- Client/Server mode, with browser program as client
- Client and server interact with HTTP: Hyper-Text Transfer Protocol(超文本传送协议)
- WWW documents are located with URL: Uniform Resource Locator(统一资源定位符)
- WWW documents are written in HTML: HyperText Markup Language(超文本标记语言)
- Searching engines facilitate user to find information needed

# Architectural Overview of WWW



# Client Side: Browser(浏览器)

- An application program, as user's interface to Web
- Fetching information (web page) from Web server
- Displaying information for users
- After the user clicks on a hyperlink, Browser
  - Determines URL, name of the linked web page, eg. *www.abc.com/example.html*
  - Asks DNS for the IP address of www.abc.com
  - Establishes a TCP connection to replied IP address
  - Sends over an HTTP request asking for the page /example.html
  - Fetches embedded images
  - Displays all the texts and images in /example.html
  - Releases TCP connection

# Web Server

- Stores set of Web documents
- Responds to request from browser by sending copy of document
- Steps of server's loop
  - Accept a TCP connection from a client (a browser).
  - Get the path to the page, which is the name of the file requested.
  - Get the file (from disk).
  - Send the contents of the file to the client.
  - Release the TCP connection

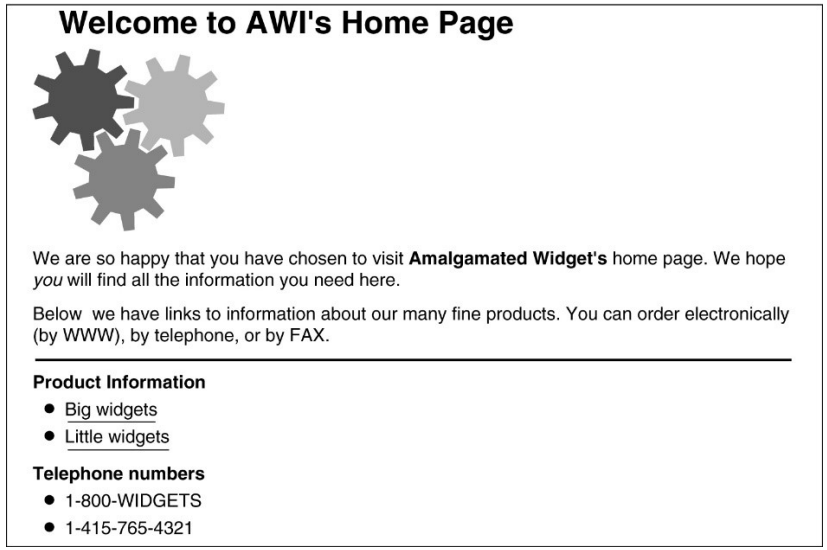
# Web Documents

- Also called a web page
- Types
  - Static web page
    - Stored in file
    - Unchanging
  - Dynamic web page
    - Can change whenever a server receives a request
    - Generated on server side
    - Output from a program on server, usually a script
  - Active web page
    - Can change after the document has been loaded into a browser
    - Generated on client side
    - Consists of a computer program, known as Java, running in browser
    - Document called applet

# HTML – 超文本标记语言

```
<html>
<head><title> AMALGAMATED  WIDGET, INC. </title> </head>
<body> <h1> Welcome to AWI's Home Page</h1>
<img src="http://www.widget.com/images/logo.gif" ALT="AWI Logo"> <br>
We are so happy that you have chosen to visit <b> Amalgamated Widget's </b>
home page. We hope <i> you </i> will find all the information you need here.
<p>Below we have links to information about our many fine products.
You can order electronically (by WWW), by telephone, or by fax. </p>
<hr>
<h2> Product information </h2>
<ul>
   <li> <a href="http://widget.com/products/big"> Big widgets </a>
   <li> <a href="http://widget.com/products/little"> Little widgets </a>
</ul>
<h2> Telephone numbers</h2>
<ul>
   <li> By telephone: 1-800-WIDGETS
   <li> By fax: 1-415-765-4321
</ul>
</body>
</html>
```

(a)

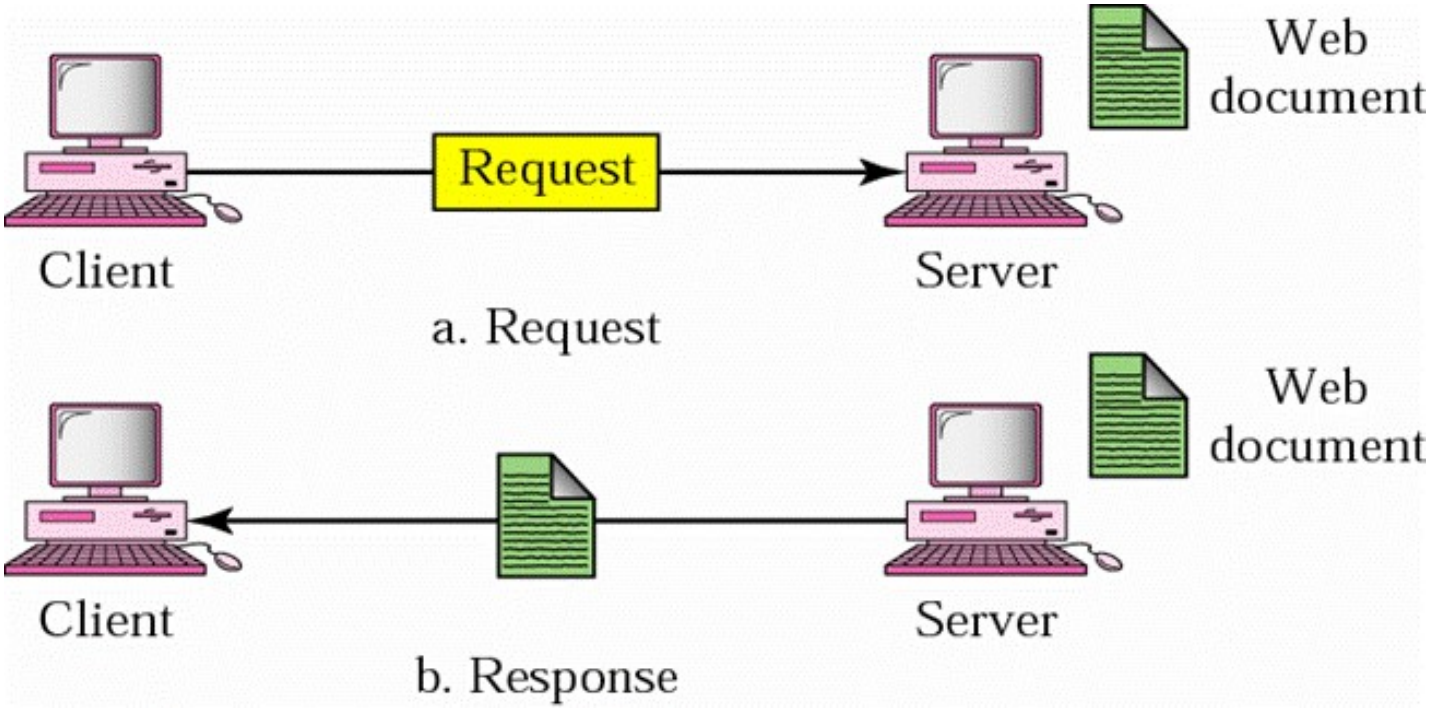



(b)

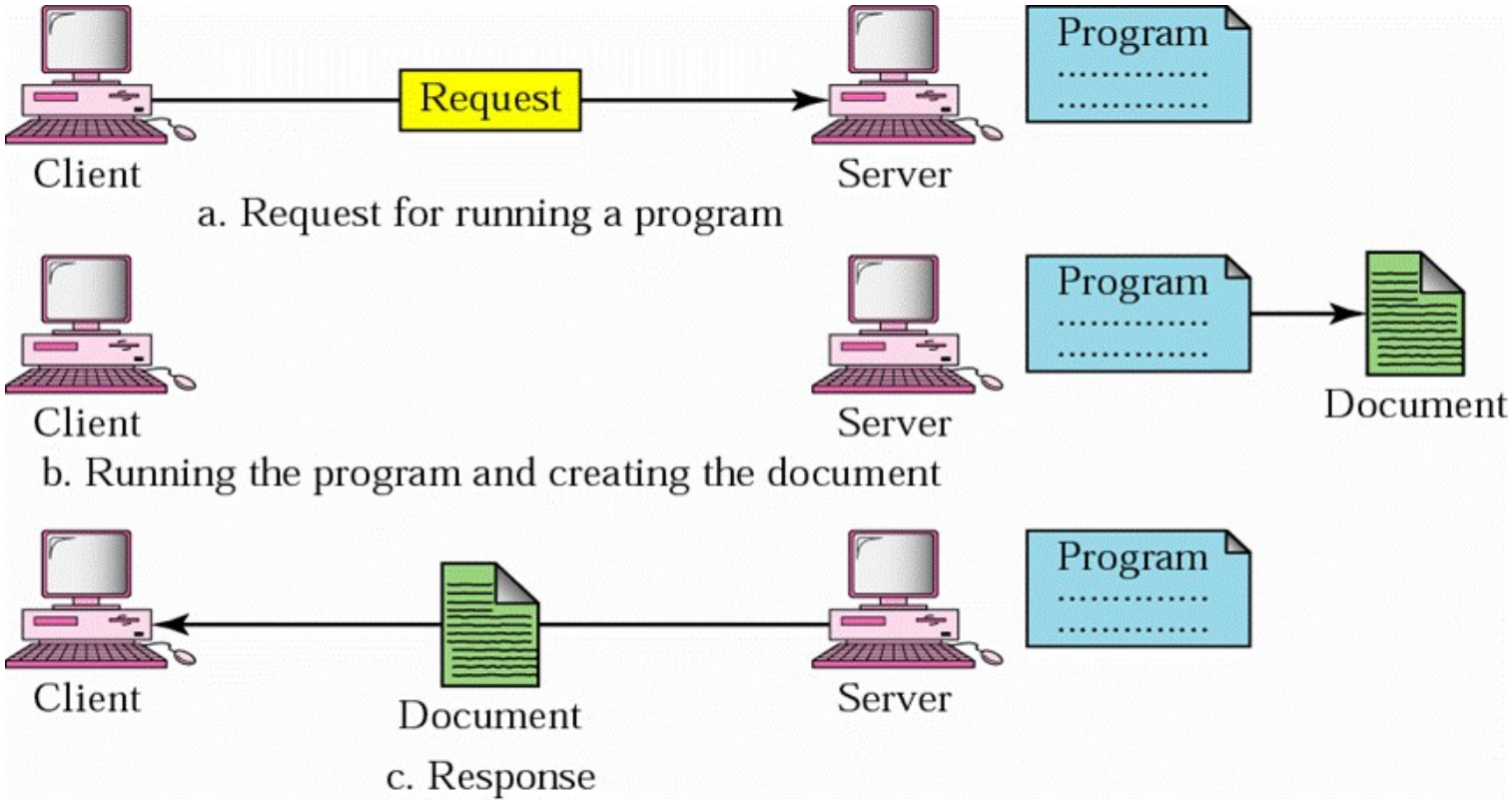
(a) 一个样例网页的超文本标记代码.

(b) 格式化网页.

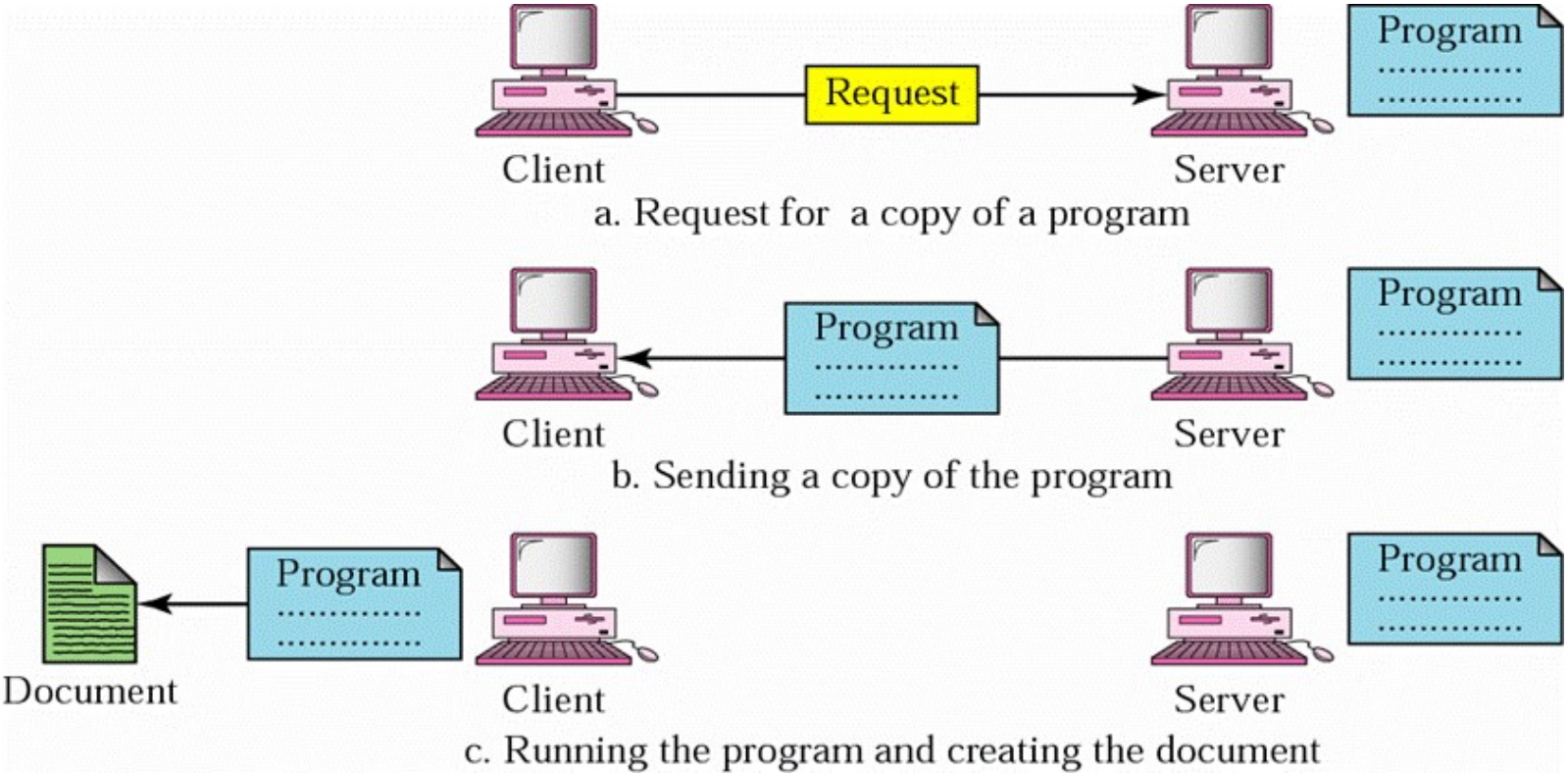
# Static web pages



# Dynamic web pages

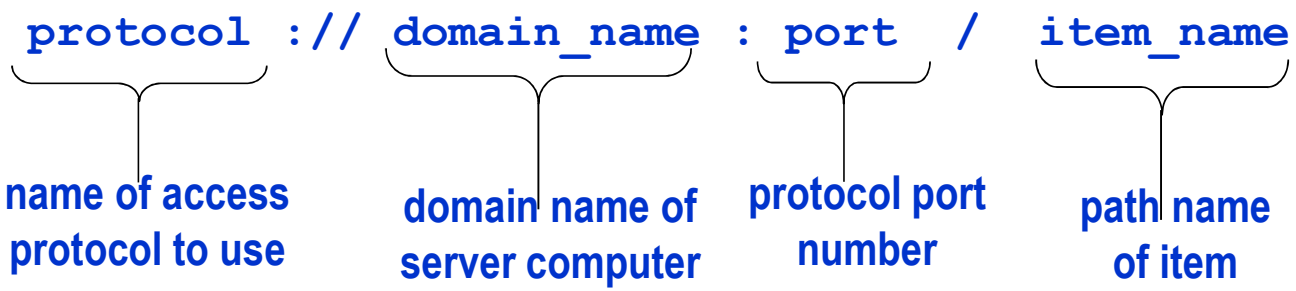


# Active Web pages



# Links to Other Pages

- Embedded in HTML document
- Browser
  - Hides text of link from user
  - Associates link with item on page
- Called Uniform Resource Locator (URL)
  - Serving as worldwide name of a web page
  - By default, Protocol is *http* ; Port is *80*; Path is *index.html*



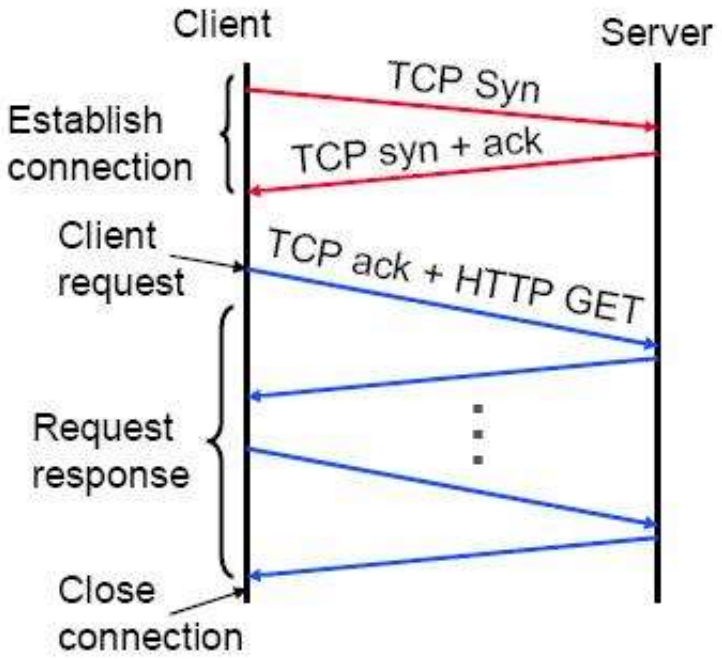
# Some Common URLs

- Web browsers are often called Universal Clients because most can talk other protocols besides HTTP

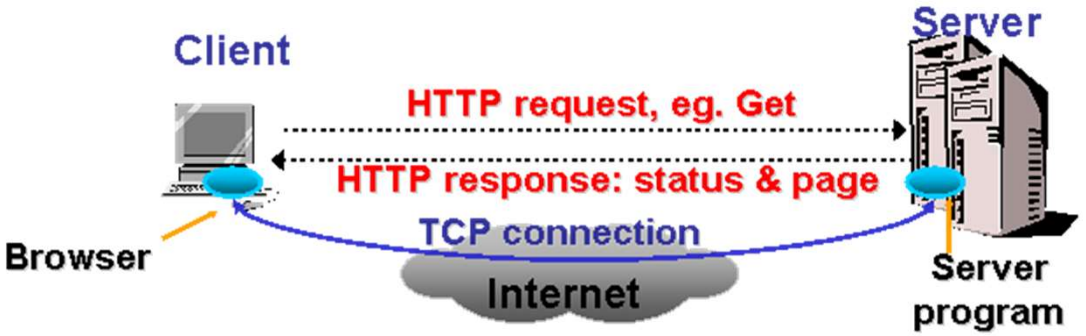
| Name | Used for | Example |
|---|---|---|
| http | Hypertext (HTML) | http://www.ee.uwa.edu/~rob/ |
| https | Hypertext with security | https://www.bank.com/accounts/ |
| ftp | FTP | ftp://ftp.cs.vu.nl/pub/minix/README |
| file | Local file | file:///usr/suzanne/prog.c |
| mailto | Sending email | mailto:JohnUser@acm.org |
| rtsp | Streaming media | rtsp://youtube.com/montypython.mpg |
| sip | Multimedia calls | sip:eve@adversary.com |
| about | Browser information | about:plugins |

# HTTP : Hyper-Text Transfer Protocol

- Client-server architecture
- Synchronous request/reply protocol
  - Running over TCP, Port 80
- Stateless
  - Each command is executed independently, without any knowledge of the previous operations



# Operations of HTTP




(1) Browser analyses URL of the hyperlink, i.e. http://www.abc.com/example.html

(2) Browser uses DNS to get IP address of the server (*www.abc.com*)

(3) Browser sets up a TCP connection with server

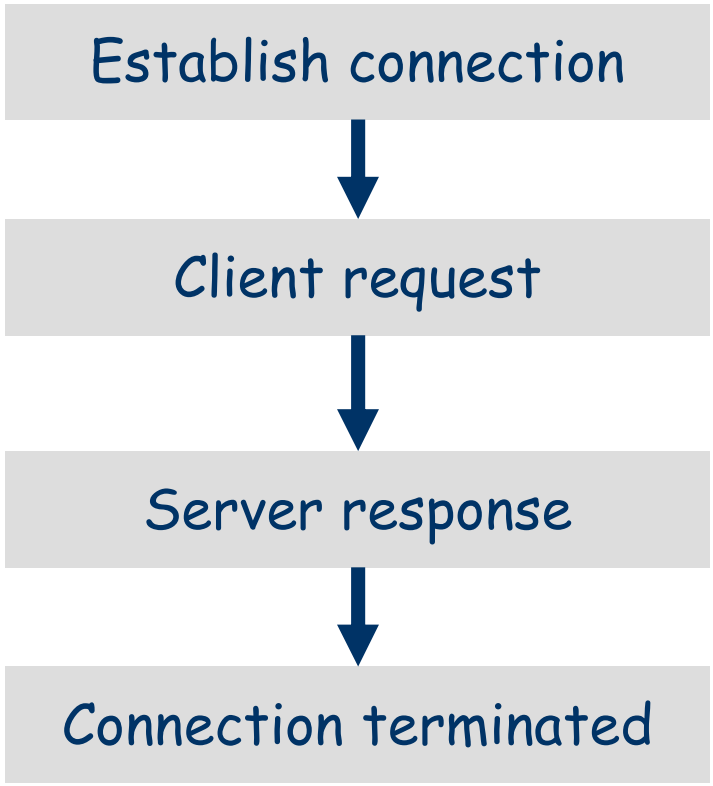
(4) Browser sends a HTTP request: GET /example.html HTTP/1.0

(5) Server sends requested page to browser

(6) TCP connection is released

(7) Browser displays example.html

# HTTP Transaction




- TCP connection set up
- uses a port number as application reference
- usually port 80

- HTTP message sent with a request line
- request-line = method URL HTTP version

- server sends HTTP message and optionally requested data
- resp-message = HTTP version status code reason-phrase [optional stuff]

- usually by the server
- sometimes the client "stops" it
- anything else, whoever notices terminates

# HTTP Methods

| Method | Description |
|---|---|
| GET | Request to read a Web page |
| HEAD | Request to read a Web page's header |
| PUT | Request to store a Web page |
| POST | Append to a named resource (e.g., a Web page) |
| DELETE | Remove the Web page |
| TRACE | Echo the incoming request |
| CONNECT | Reserved for future use |
| OPTIONS | Query certain options |

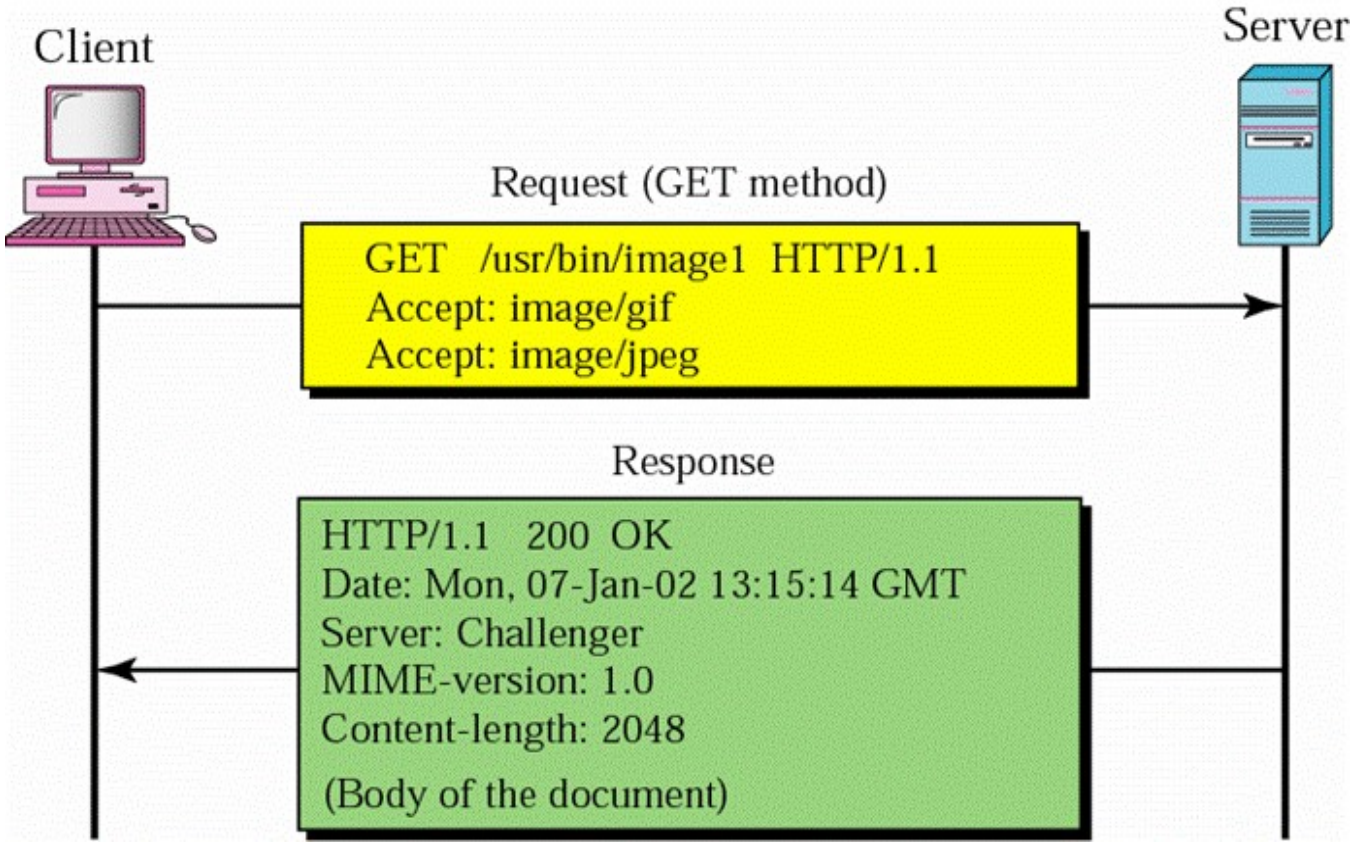
# HTTP Message Headers

| Header | Type | Contents |
|---|---|---|
| User-Agent | Request | Information about the browser and its platform |
| Accept | Request | The type of pages the client can handle |
| Accept-Charset | Request | The character sets that are acceptable to the client |
| Accept-Encoding | Request | The page encodings the client can handle |
| Accept-Language | Request | The natural languages the client can handle |
| Host | Request | The server's DNS name |
| Authorization | Request | A list of the client's credentials |
| Cookie | Request | Sends a previously set cookie back to the server |
| Date | Both | Date and time the message was sent |
| Upgrade | Both | The protocol the sender wants to switch to |
| Server | Response | Information about the server |
| Content-Encoding | Response | How the content is encoded (e.g., gzip) |
| Content-Language | Response | The natural language used in the page |
| Content-Length | Response | The page's length in bytes |
| Content-Type | Response | The page's MIME type |
| Last-Modified | Response | Time and date the page was last changed |
| Location | Response | A command to the client to send its request elsewhere |
| Accept-Ranges | Response | The server will accept byte range requests |
| Set-Cookie | Response | The server wants the client to save a cookie |

# HTTP Status Codes

| Code | Meaning | Examples |
|---|---|---|
| 1xx | Information | 100 = server agrees to handle client's request |
| 2xx | Success | 200 = request succeeded; 204 = no content present |
| 3xx | Redirection | 301 = page moved; 304 = cached page still valid |
| 4xx | Client error | 403 = forbidden page; 404 = page not found |
| 5xx | Server error | 500 = internal server error; 503 = try again later |

# Example of HTTP Transaction



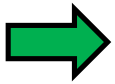

*[Forouzan]*

# HTTP/1.1 Performance Enhancements

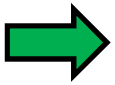
- HTTP/1.0 is a "stop and wait" protocol
  - Separate TCP connection for each file
    - Connect setup and tear down is incurred for each file
    - Inefficient use of packets
    - Server must maintain many connections
- HTTP/1.1 specification focus on performance enhancements
  - Persistent connections
  - Pipelining
  - Enhanced caching options
  - Support for compression

# HTTP/1.1
# Persistent Connections and Pipelining

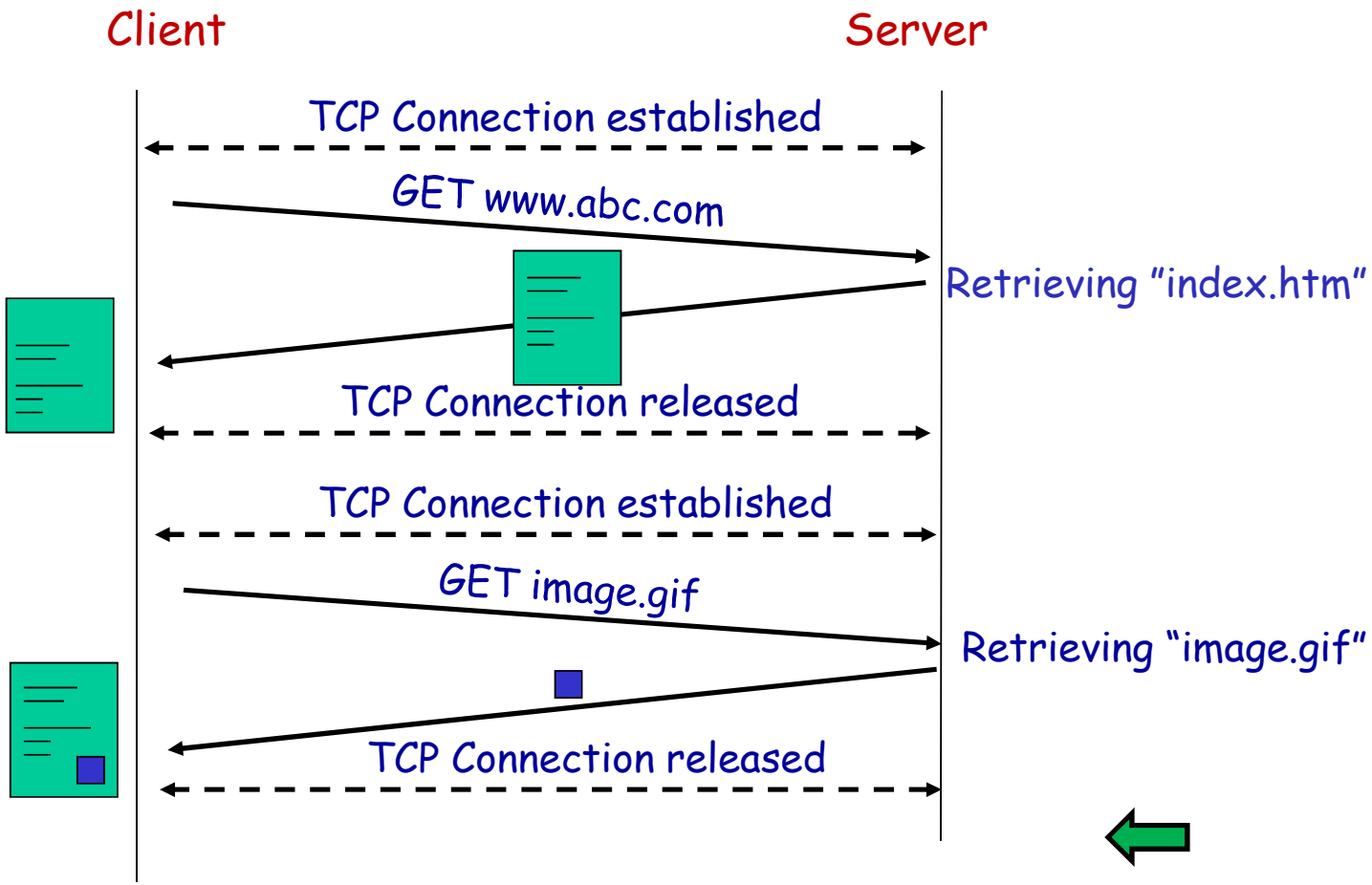
- Persistent connections(持续连接)
  - Use the same TCP connection(s) for transfer of multiple files
  - Reduces packet traffic significantly



- Pipelining(流水线)
  - Multiple HTTP requests can be written out to a socket together without waiting for the corresponding responses.
  - Pack several HTTP requests into one TCP/IP packet

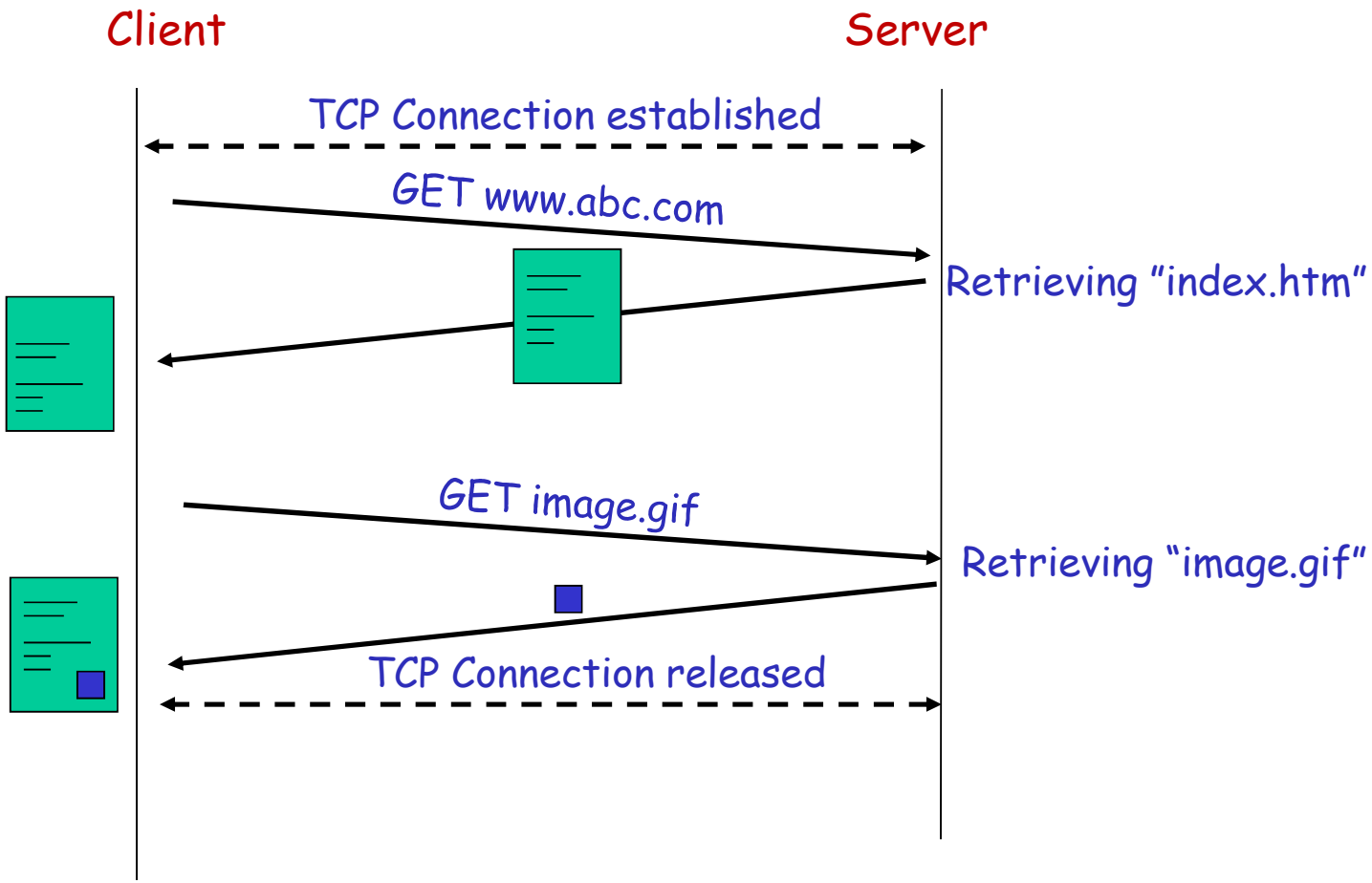


# Example of Non-persistant connections



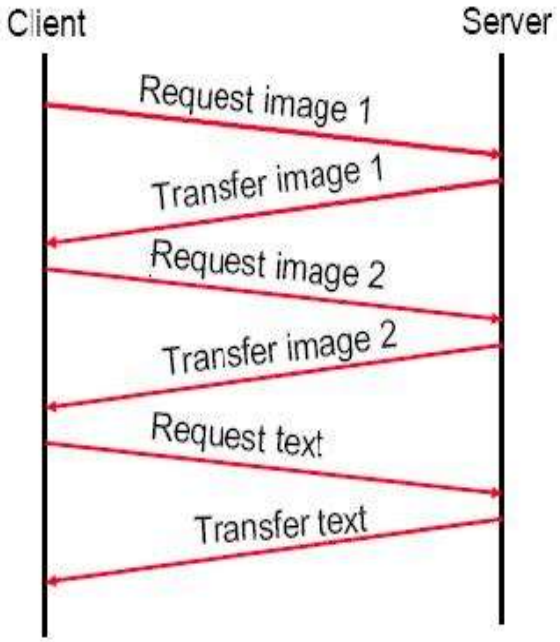

*[Kurose]*

# Example of Persistent connections

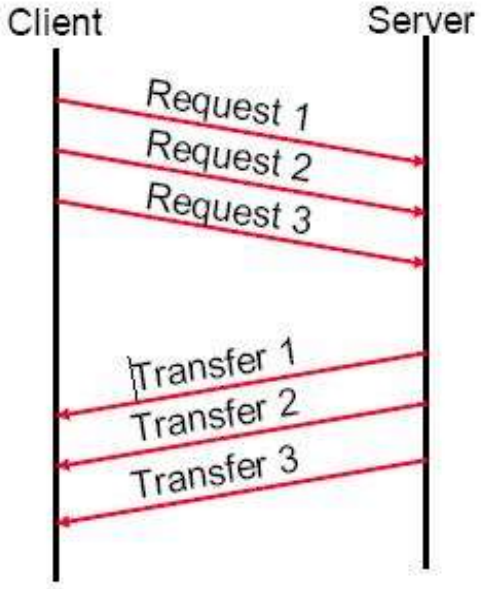



*[Kurose]*

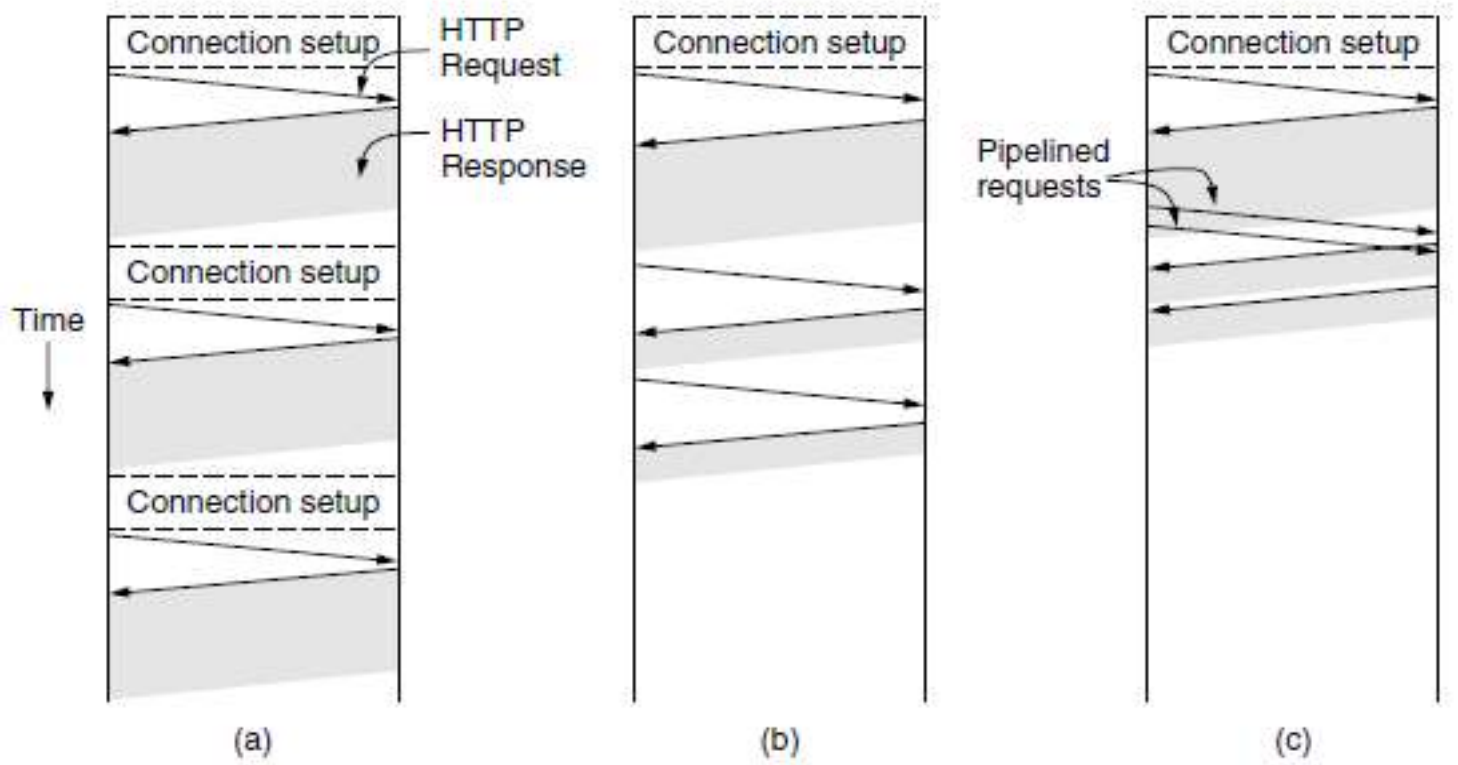
# Example of Pipelining(流水线)




Non-pipelining




Pipelining




multiple connections and sequential requests

A persistent connection and sequential requests

A persistent connection and pipelined requests

# User-server state: cookies

- Many major Web sites use cookies

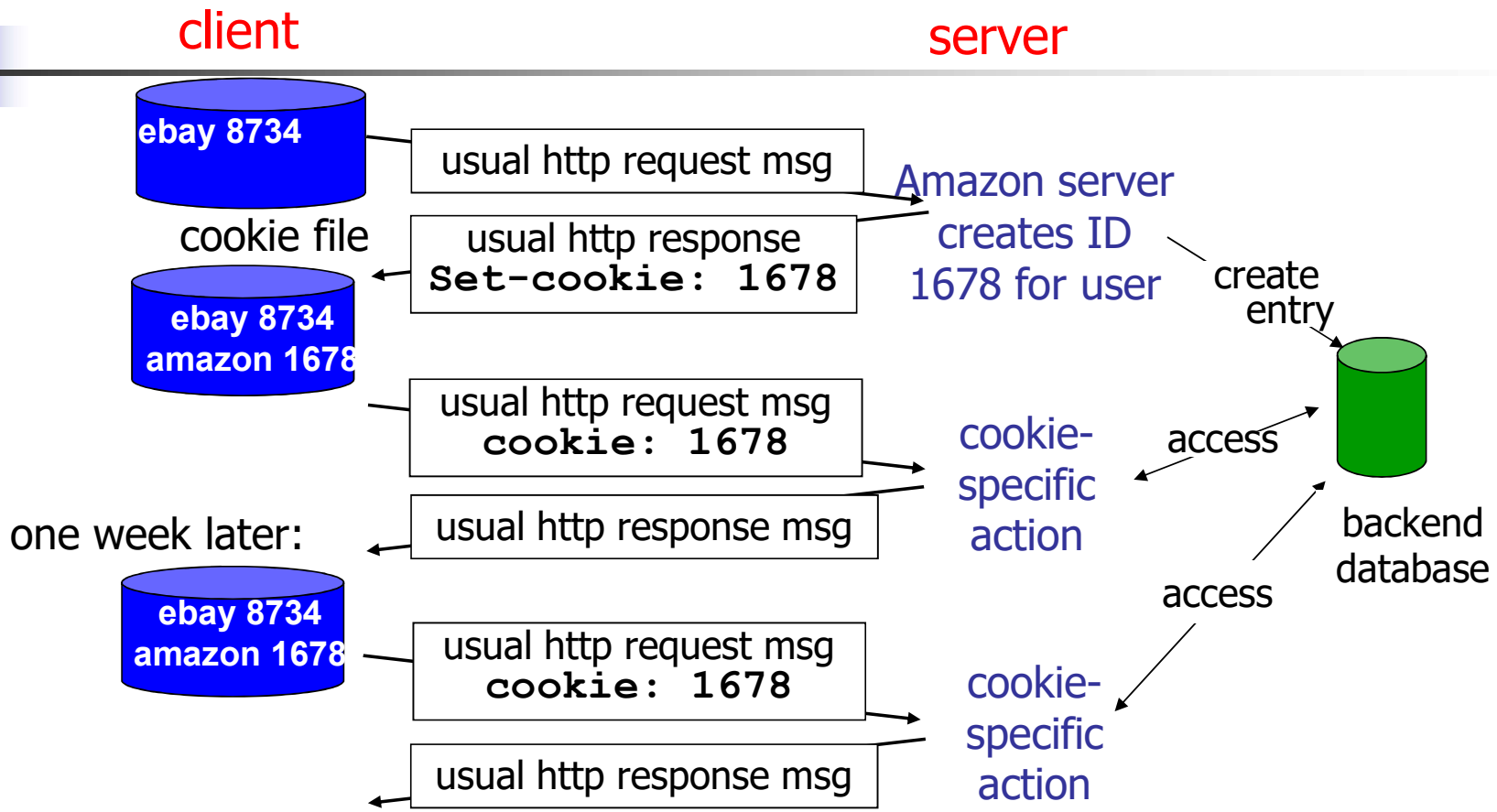
Example:

- Susan always access Internet from PC
- She visits specific e-commerce site for first time
- when initial HTTP request arrives at site, site creates:
  - unique ID
  - entry in backend database for ID

Four components:

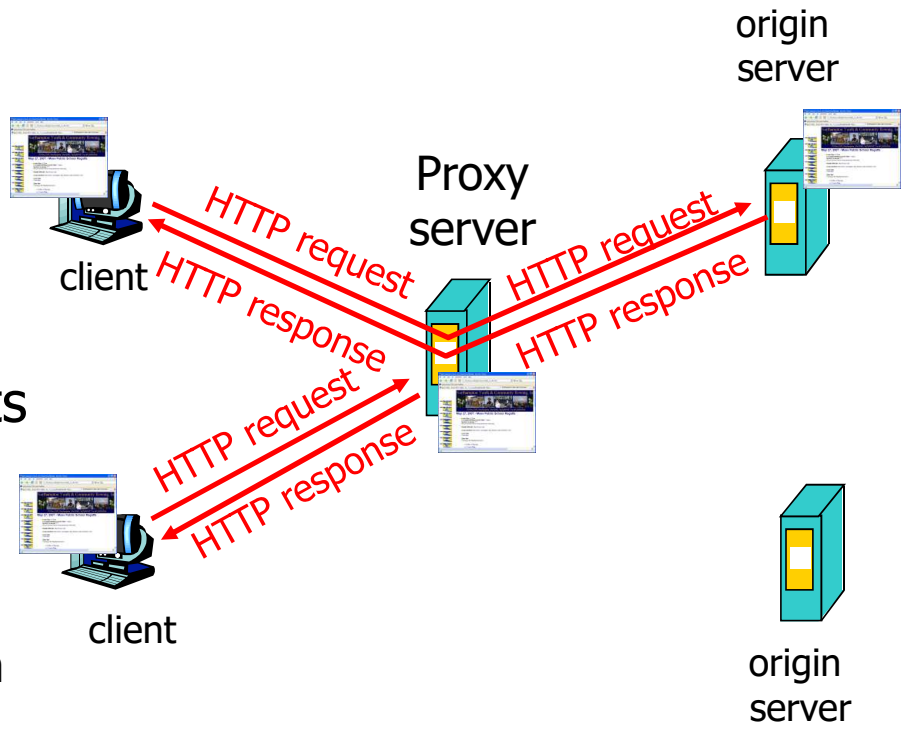
1) cookie header line of HTTP *response* message
2) cookie header line in HTTP *request* message
3) cookie file kept on user's host, managed by user's browser
4) back-end database at Web site

# Cookies: keeping "state"



# Web Caches (Proxy Server)

- Motivation: satisfy client request without involving origin server
- User sets browser: Web accesses via a proxy server
- Browser sends all HTTP requests to proxy server
  - If requested file in cache: proxy server returns file
  - else proxy requests file from origin server, then forwards to client

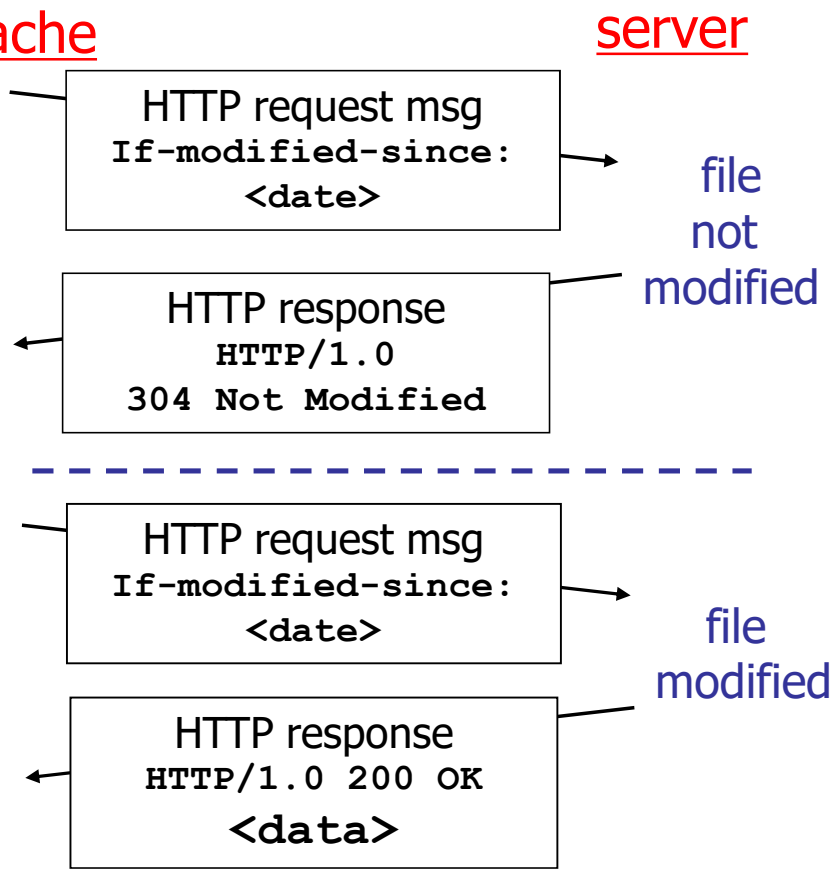


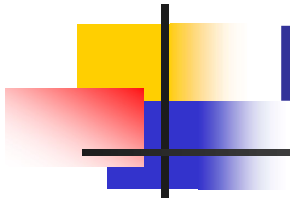
# Conditional get

- Server does not send required files if cache has up-to-date cached version
- cache: specify date of cached copy in HTTP request

  `If-modified-since: <date>`
- server: response contains no object if cached copy is up-to-date:

  `HTTP/1.0 304 Not Modified`





# HTTPS

- Hypertext Transfer Protocol Secure (HTTPS)
- HTTP on top of the SSL(Secure Sockets Layer)/TLS (Transport Layer Security) protocol
- A communication protocol for secure communication over a computer network, with especially wide deployment on the Internet

# WWW小结

- 目标：将分布在互联网不同设备上的资源组织起来。
- WWW体系结构
  - 超链接：超文本、超媒体
  - Web客户(浏览器)和Web服务器
  - URL：protocol://domain_name:port/item_name
- Web页面：静态页面(HTML)，动态页面，活动页面
- 超文本传输协议(HTTP)
  - Client/Server, 同步数据传输
  - over TCP, port 80
  - HTTP操作：TCP连接，HTTP请求报文和响应报文
  - HTTP1.0&1.1: 持久连接、流水线请求

# Key Points(1)

- Idea of client/server model
- DNS
  - Domain name, hierarchical architecture
  - Jobs of resolver and DNS servers
  - Iterative resolution vs. Recursive resolution
- WWW
  - Concepts of WWW, HTTP, HTML, hyperlink, URL, browser
  - Operations of HTTP
  - HTTP 1.0 vs. HTTP 1.1
  - Function of cookies, proxy and conditional get

# Key Points(2)

- Email
  - Email address
  - SMTP: sending Email from user machine to receiver's mailbox
  - POP3/IMAP: downloading Email from mailbox to user machine
  - Web mail: using HTTP

# Terms

| 缩写 | 英文全称 | 中文 |
|---|---|---|
| | Application Layer | 应用层 |
| | Authoritative Name Server | 权威名字服务器 |
| | browser | 浏览器 |
| | distributed database | 分布式数据库 |
| | domain | 域 |
| DNS | Domain Name System | 域名系统 |
| | hyperlink | 超链接 |
| HTML | HyperText Markup Language | 超文本标记语言 |
| HTTP | HyperText Transfer Protocol | 超文本传送协议 |
| IMAP | Internet Message Access Protocol | 因特网报文访问协议 |
| | Internet Message Format | 因特网报文格式 |
| | Iterative resolution | 迭代查询 |
| MIME | Multipurpose Internet Mail Extension | 多用途邮件扩展 |
| MTA | Mail Transfer Agent | 邮件传送代理 |

| 缩写 | 英文全称 | 中文 |
| --- | --- | --- |
| | Name Server | 名字服务器 |
| | Name space | 名字空间 |
| | Non-Persistent connections | 非持久连接 |
| P2P | Peer-to-peer | 对等体 |
| | Persistent connections | 持久连接 |
| | pipelining | 流水线 |
| POP | Post Office Protocol | 邮局协议 |
| | proxy | 代理 |
| | Recursive resolution | 递归解析 |
| | resolver | 解析器 |
| | Resource records | 资源记录 |
| | searching engine | 搜索引擎 |
| SMTP | Simple Mail Transfer Protocol | 简单邮件传送协议 |

| 缩写 | 英文全称 | 中文 |
| --- | --- | --- |
| SMTP | Simple Mail Transfer Protocol | 简单邮件传送协议 |
| | User Agent | 用户代理 |
| URL | Uniform Resource Locator | 统一资源定位符 |
| WWW | World Wide Web | 万维网 |
| | zone | 区 |

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  - *"计算机网络", 谢希仁，第五版，电子工业出版社，2008年，which are marked with [谢]*